



**DORSU ALUMNI TRACER: A DATA-DRIVEN APPROACH TO MONITORING
ALUMNI CAREER PROGRESS**

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**A BSIT Capstone Project Submitted to the Faculty of Computing, Engineering,
and Technology of the Davao Oriental State University in Partial for the Fulfillment
Requirements for the Degree**

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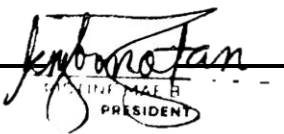


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The Researchers

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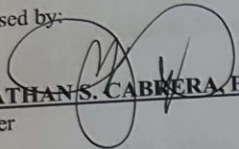


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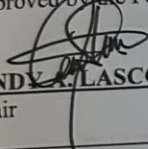
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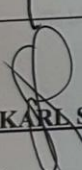

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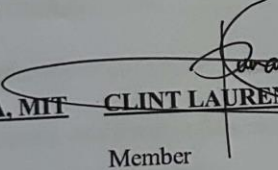
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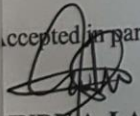

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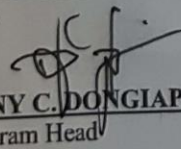

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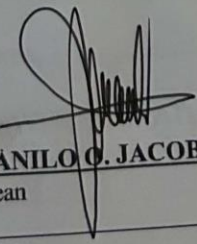
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TABLE OF CONTENTS

TITLE	PAGE
Title Page	i
Academic Integrity Declaration	ii
Acknowledgement	iii
Approval Sheet.....	iv
Table of Contents	v
List of Tables	vi
List of Figures.....	vii
List of Appendices	viii
Abstract	xi
CHAPTER I – INTRODUCTION	1
1.1. Rationale of the Study	1
1.2. Purpose and Project Description	2
1.3. Objectives of the Study	3
1.4. Significance of the Study.....	4
1.5 Scope and Limitation.....	5
1.6. Conceptual Framework.....	6
1.7 Definition of Terms	7
CHAPTER II – REVIEW RELATED LITERATURE	8
2.1. Literature Map.....	8
2.2. Related Literature.....	9

	vi
2.3. Synthesis.....	12
2.4. Related System.....	13
CHAPTER III – MATERIALS AND METHODS	19
3.1. Software Methodology	19
3.2. System Planning	20
3.3. System Analysis.....	28
3.4. System Design	32
3.5. System Implementation.....	42
3.6. System Maintenance.....	44
CHAPTER IV – RESULTS AND DISCUSSIONS.....	46
CHAPTER V – SUMMARY, CONCLUSION, AND RECOMMENDATIONS	70
5.1. Summary.....	71
5.2. Conclusion	71
5.3. Recommendation	71
LITERATURE CITED	72
CURRICULUM VITAE	141

LIST OF TABLES

TABLE	PAGE
Table 1. Tabulated Comparison	18
Table 2. Cost–Benefit Analysis	23
Table 3. Technical Feasibility	25
Table 4. Admin.....	34
Table 5. Alumni	35
Table 6. Program Head	35
Table 7. Alumni Survey Form	36
Table 8. Admin Evaluation Form Result.....	56
Table 9. Alumni Evaluation Form Result	59

LIST OF FIGURES

FIGURE	PAGE
Figure 1. Conceptual Framework of DOrSU Alumni Tracer System	6
Figure 2. Literature Map	8
Figure 3. Online Alumni Tracer System for Cavite State Graduates	12
Figure 4. PUP eAlumni Portal	13
Figure 5. Alumni Tracking System for Aemilianum College Inc	14
Figure 6. La Salle University–Ozamiz (LSU) eTracer.....	15
Figure 7. Modified Waterfall Cycle.....	19
Figure 8. Work Breakdown Structure.....	26
Figure 9. GANTT CHART.....	27
Figure 10. BPMN of Current System	28
Figure 11. System Architecture.....	33
Figure 12. Entity Relationship Diagram	34
Figure 13. Level 0 DFD	37
Figure 14. Level 1 DFD	38
Figure 15. Use Case Diagram	39
Figure 16. UX Sitemap	40
Figure 17. Alumni Registration.....	45
Figure 18. Alumni Profiling	45
Figure 19. Collecting alumni employment information.....	47
Figure 20. Alumni Dashboard	48
Figure 21. Program Head Dashboard.....	48
Figure 22. Admin Dashboard	48
Figure 23. Employment Trend.....	49

Figure 24. Employment after Graduation	49
Figure 25. Visualize real-time employment patterns.....	51
Figure 26. Matching algorithms for advanced analysis	52
Figure 27. Personalized dashboards and insights	54
Figure 28. Alumni Tracer Report.....	55

LIST OF APPENDICES

APPENDIX	PAGE
Appendix A. Letter of request and approval to conduct the study	74
Appendix B. Routing Slip	76
Appendix C. Grammarly Report	78
Appendix D. Grammarian's Certification	80
Appendix E. Expert Evaluation Form.....	82
Appendix F. Survey User Evaluation Questionnaire.....	97
Appendix G. Sample filled-out User Evaluation Questionnaires	116
Appendix H. Statistician's Certificate	132
Appendix H. User Testing Photo Documentation	134
Appendix I. DOrSU Alumni Tracer User Manual	136

ABSTRACT

Kristine Mae Bonotan, Mike Misoles, and Kenneth Jay Paragoso “DOrSU Alumni Tracer: A Data-Driven Approach to Monitoring Alumni Career Progress” (BSIT Capstone Project). Davao Oriental State University, June 2026.

Adviser: **Jonathan Cabrera**

The DOrSU Alumni Tracer System is a comprehensive web-based platform developed to modernize graduate tracking at Davao Oriental State University. This research addresses the challenges of manual alumni monitoring by implementing a digital solution that efficiently collects, analyzes, and manages graduate career data. The system aims to: (1) establish a secure platform for alumni registration and profile management, (2) implement digital data collection for employment information, (3) develop rule-based access controls for different user types, (4) integrate data analytics to identify employment trends, and (5) create interactive dashboards for data visualization. The development of the system followed the Software Development Life Cycle (SDLC), which provided a structured approach through the phases of planning, analysis, design, development, testing, deployment, and maintenance to ensure the platform's quality, reliability, and effectiveness. Through data analytics and decision tree algorithms, the system identifies employment trends and assesses job-degree relevance, while interactive dashboards provide real-time visualization of employment patterns. The platform's responsive design ensures accessibility across various devices, supported by a cloud-based infrastructure for security and scalability. Implementation results demonstrate significant improvements in data accuracy, alumni engagement, and institutional decision-making. The system successfully automates report generation for accreditation purposes and provides actionable insights for curriculum enhancement. This project confirms that the digital transformation of alumni tracer systems can substantially improve institutional effectiveness, strengthen alumni-university relationships, and support data-driven academic planning in higher education institutions.

Keywords: Alumni Tracer System, Web-based Platform, Graduate Tracking, Data Analytics, Rule-based Algorithm, Institutional Planning

CHAPTER I

INTRODUCTION

1.1 Rationale of the Study

Globally, higher education institutions encounter significant challenges in accurately monitoring graduates' career outcomes due to fragmented and manual tracking methods. These limitations result in delays, incomplete data, and difficulties in aligning academic programs with labor market demands. In the Philippines, most universities depend on tracer studies that frequently utilize manual questionnaires and outdated alumni records, leading to low response rates and unreliable information for curriculum enhancement and policy development (CHED, 2019). At the local level, Davao Oriental State University (DOrSU) experiences similar challenges, as its manual data collection process is susceptible to errors, inefficiencies, and incomplete records. These issues restrict the university's capacity to assess program effectiveness, identify skill gaps, and adapt academic offerings to meet changing labor market requirements.

Maintaining an orderly and comprehensive record of graduates' career trajectories remained challenging for many universities. Most institutions relied on traditional methods such as informal communication or paper-based surveys, which frequently led to low response rates, inconsistent data, and prolonged processing times. These limitations hindered the collection of relevant information that correctly reflected whether academic programs supported graduates' success in their respective careers or entrepreneurial pursuits. Studies on tracer exercises indicated that response rates frequently fell below 50%, mainly due to outdated contact information, survey fatigue, and the labor-intensive nature of manual data collection (Garcia, 2021; World Bank, 2020). Tracer studies served as vital instruments for assessing graduate employability and the alignment of academic programs with labor employment market demands, as noted by Luciano et al. (2020).

However, many institutions continued to lack a productive framework for managing this process.

In order to gather, arrange, and evaluate alumni data more precisely and effectively, the DOrSU Alumni Tracer: A Data-Driven Approach to Monitoring Alumni Career Progress project proposed a web-based system. The goal of this project was to use data visualization and analytics to determine employment trends, job relevancy, and the geographic distribution of graduates. Through the system, DOrSU was able to enhance institutional planning, alumni engagement, academic program evaluation, and decision-making. Consequently, this study helped improve data management, boost alumni engagement, and strengthen closer ties between workforce development and education.

1.2 Purpose and Project Description

The project monitored and analyzed the career progression of Davao Oriental State University (DOrSU) graduates. The implemented system addressed inefficiencies in the previous manual tracer process by utilizing information technology to collect accurate data, enhance alumni engagement, and provide meaningful visual insights. Through the use of structured digital forms and engagement tools, the system collected key information, including employment status, job relevance, industry sector, and geographic location.

The dashboard feature of the system utilized an easy-to-use interface, allowing users to track real-time information about employment using various forms of graphical and charted displays. The system also had the ability to determine what was the relationship between the major/program that students graduated from and where they are currently working based on the combination of a rules-based algorithm along with job course-matching logic. In order to encourage active engagement by alumni, the system provided each alumnus or alumna with personalized career advice, visualizations of their

professional development over time, and opportunities to grow their connection to the university community. The system also included many other convenient features, including automated reminders, alumni/alumni association feedback options, and a mobile-friendly design. The system contained a central cloud database to store all data collected and was designed with role-based access controls (Alumni, Admin, Program Head).

3. Objectives of the Study

This project aims to:

Develop and implement a secure web-based DOrSU Alumni Tracer System that effectively monitors and analyzes graduate career progression.

Specifically, this project aims to:

- Build a secure web-based system where alumni could register and update personal and employment details
- Create digital forms that collect data such as job title, industry sector, employment status, and job relevance
- Design role-based access controls for Alumni, Admin, and Program Heads
- Integrate data analytics that identify trends in employment
- Develop interactive dashboards and charts that visualize employment patterns in real-time
- Apply rule-based algorithms to job-course matching
- Launch a fully responsive system accessible via web platforms
- Provide personalized dashboards and insights for university administrators and program evaluators
- Improve alumni data accuracy and produce at least two institutional reports annually

User Evaluation (based on ISO 25010 standards)

The tracer system is evaluated using an internationally recognized standard (ISO/IEC 25010) to determine how well it functions in terms of functionality, performance, usability, reliability, maintainability, and security; eligible users will utilize the system and evaluate its usefulness as a tool for promoting language acquisition.

1.4 Significance of the Study

The DOrSU Alumni Tracer System positively affects the community by supporting the production of graduates with skills relevant to local industry needs. This enables businesses to source qualified employees, strengthens ties between the university and the workforce, and fosters a greater sense of community among alumni that may lead to mentoring relationships and volunteer opportunities, which could positively impact the overall development of the community.

Alumni. Graduates may use the system to update their employment and career information, stay connected with the university, and access job-related resources and opportunities.

DOrSU Administrators. The system provides administrators with valuable data and insights regarding graduate employment trends. These insights assist the university in evaluating and improving academic programs, ensuring that they remain relevant and responsive to industry needs.

Program Heads. Program heads can track the employment outcomes of their graduates and assess how effectively their academic programs prepare students for the workforce. This information supports continuous program improvement and helps maintain the quality and relevance of the university's educational offerings.

1.5 Scope and Limitations

The scope of this study focuses on the development and validation of the DOrSU Alumni Tracer System, a secure web-based platform designed to facilitate the efficient collection, management, and analysis of alumni information.

- Development of a centralized web-based alumni tracer system for DOrSU graduates
- Implementation of automated data collection through digital forms and online surveys
- Design of role-based access for three user types: alumni, administrators, and program heads
- Creation of interactive dashboards and visualization tools for data analysis
- Integration of data analytics for tracking employment trends and career progression
- Deployment of cloud-based infrastructure with secure data storage
- Focus on collecting career-related data, including employment status, job roles, and industry sectors
- Provision of multi-platform accessibility via web and mobile browsers
- Configuration of automated reporting features for institutional use

While the system addressed the operational needs of DOrSU Alumni Tracer, the following limitations were acknowledged:

- Public users could only access aggregated data without individual records
- Data accuracy relied entirely on self-reported information from alumni
- The platform excluded job placement and recruitment services
- All features required continuous internet connectivity
- Analytics quality depended on user participation levels

- Security measures were limited to software-based protocols
- Career analysis used basic decision tree algorithms without advanced AI
- The system did not incorporate predictive analytics capabilities
- Reporting features were constrained by data completeness
- No offline functionality was available for any system features

1.6 Conceptual Framework

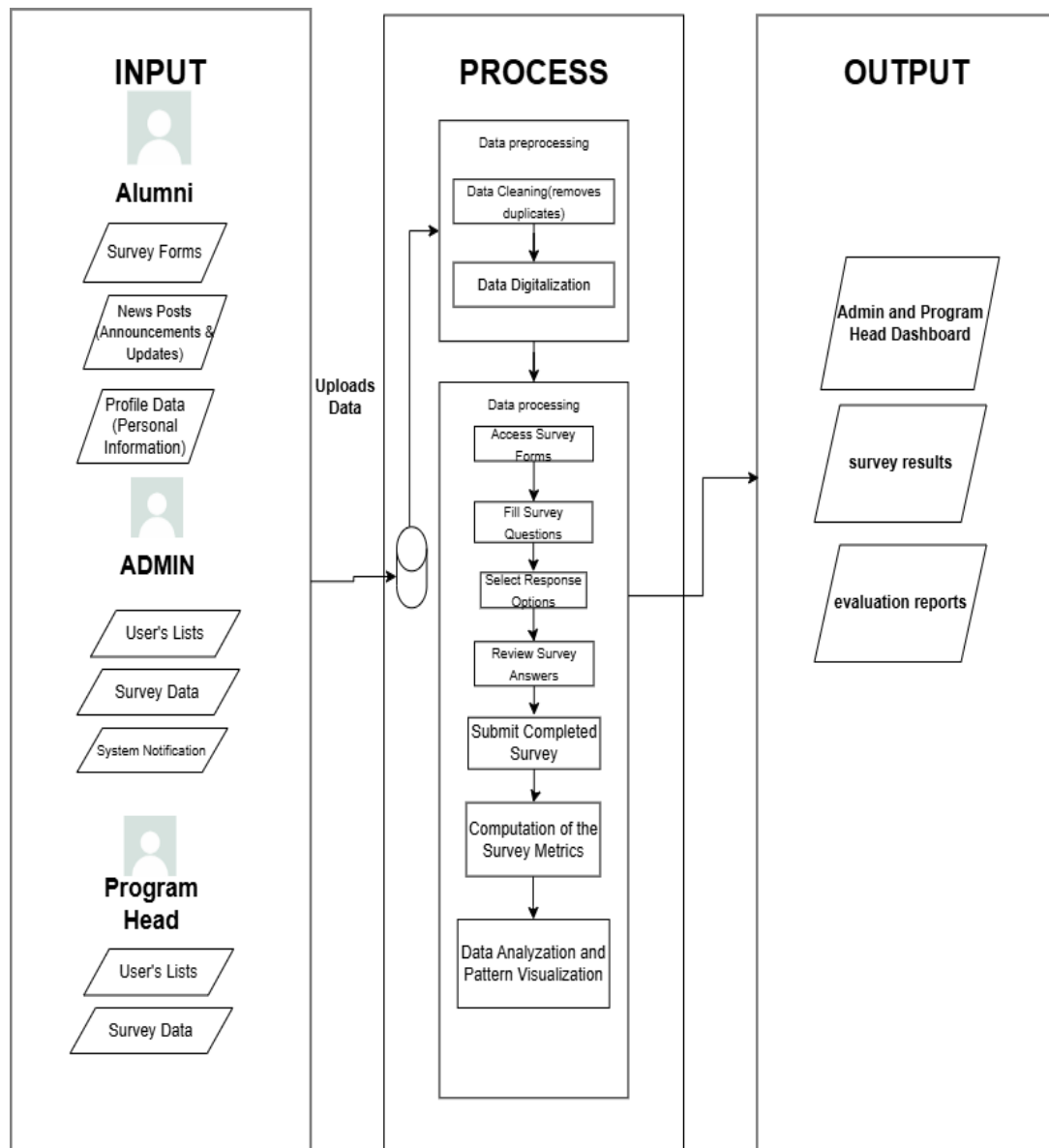


Figure 1. Conceptual Framework of DOrSU Alumni Tracer System

The conceptual model outlines how the proposed DOrSU Alumni Tracer System uses the IPO model to collect, process/transform alumni data to create useful/meaningful/alumni data-related information. The conceptual model depicts how data collected by users of the system (alumni, administrators, program heads, etc.) are transformed (preprocessed, analyzed) via a variety of methods (survey management, digitalization, data analysis) into outputs that can be used for decision-making or program evaluations. The conceptual model also portrays how the processed data is utilized (via dashboards, survey results, evaluation reports) to aid program heads/administrators in evaluating/monitoring outcomes of graduates, determining the effectiveness of programs, and contributing to an ongoing process of improving their institutions.

1.7 Definition of Terms

Alumni Tracer System

Beginning with the fact that the web-based application was developed to track and analyze graduate employment data for use by universities when making decisions about their programs.

Data Analytics

Identifying trends and measuring program performance based on data collected from alumni.

Decision

Algorithm

Tree Using a machine-learning approach to classify alumni data as well as to identify career paths.

Data Visualization

Displaying alumni data in graphical format (i.e., charts) and in dashboard format to provide an easier way to interpret the data.

Employment Status

One of the most important metrics used to determine if a graduate is employed, unemployed, self-employed, or pursuing additional education.

Data Collection

Forms and surveys created electronically that allow for the collection of structured employment-related alumni data.

Public Accessibility

An ability to display publicly aggregated data while protecting the identity of each alumni member.

Alumni Verification

Verifying graduate status utilizing student credentials, such as student ID numbers, etc., as well as verification via the institution's registrar.

CHAPTER II

REVIEW OF RELATED LITERATURE

2.1 Literature Map

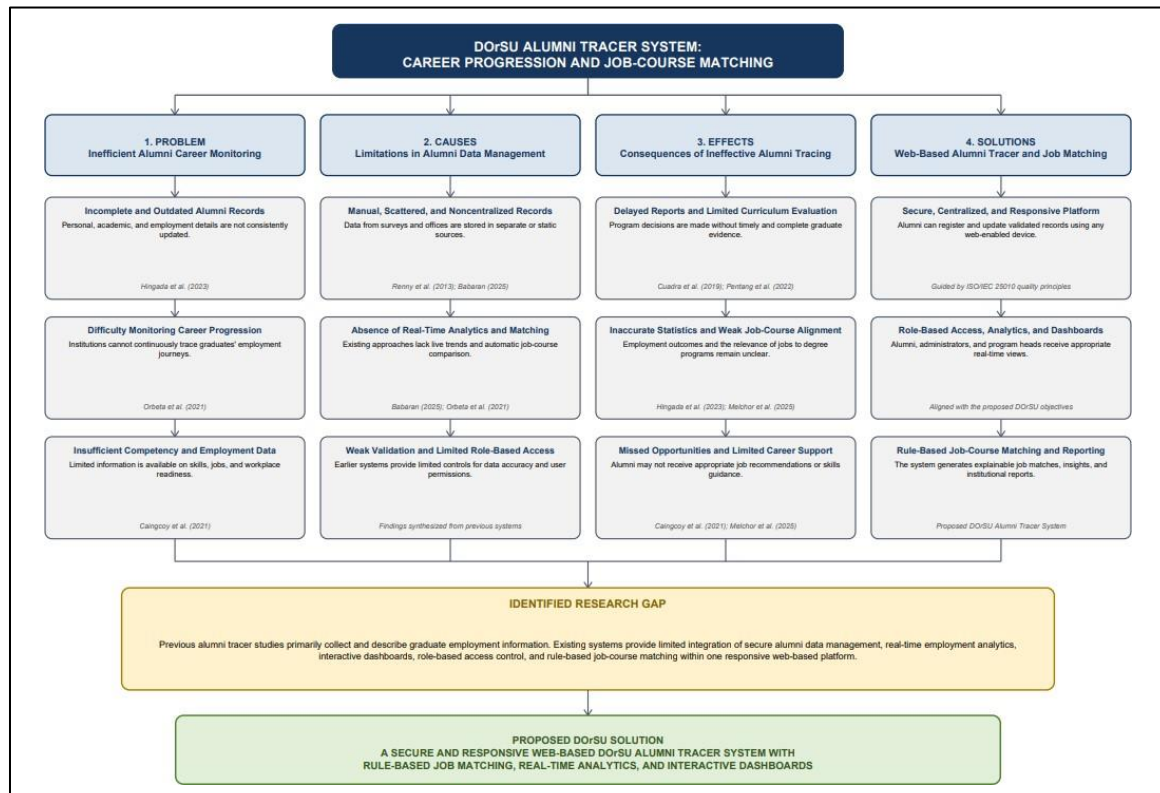


Figure 2. Literature Map

2.2 Related Literature

Alumni Tracer Studies are used by higher education institutions across the world to evaluate how well a particular Academic Program is preparing Students in terms of Labor Market Demands. Also, they use this method to measure how Employable Graduates are and to assist in Institutional Planning. Alumni Tracer Systems have been used to track various aspects of graduates' employment status, career progression, job relevance, and Satisfaction with Educational Experience. The data collected from Alumni Tracer Systems provides Universities with Data to evaluate if their Programs have adequately prepared their Students for Labor Market Demands. Additionally, this system allows for the

identification of possible Curriculum Improvements. As stated by Orozco (2018), Graduate Tracer Studies are Critical Tools that allow Institutions to Evaluate Educational Outcomes and Develop Evidence-Based Strategies for Institutional Development.

The large number of institutions continuing to struggle with the maintenance of alumni records is a major obstacle for them in maintaining this type of data. Tracer study methods are typically based on manually collecting survey responses via paper forms, which then have to be hand-entered into computers or databases. This results in an abundance of missing data points as well as lower-than-desired survey response rates. Research conducted by Sarmiento (2024) identified commonalities among educational institutions where they struggled to maintain complete and current lists of alumni due to the lack of digital platforms used for storing and updating alumni data. In addition, research findings made by Taguian and Sarbas (2021) stated that the practice of using multiple different data sources to store alumni information creates inconsistencies in how it can be analyzed and interpreted.

Web-based survey systems have been implemented as an alternative means of collecting and processing data efficiently. Web-based multi-section survey forms are increasingly being adopted by all types of institutions as a tool for gathering detailed and expansive amounts of information about their graduates' academic backgrounds, job-related experience, career development, and level of satisfaction with their education. In line with this rationale, the Commission on Higher Education (CHED) mandates that graduate tracer surveys collect several aspects of alumnus performance in order to provide institutions with information needed to evaluate the effectiveness of programs or services offered at the institution.

A further important part of a modern trace system is an alumni profile database for the central collection of all alumni information. Centralized relational database technology,

such as PostgreSQL and MySQL, offers an efficient means of collecting, storing, retrieving, and managing data on alumni. Sarmiento (2024) pointed out that the use of centralized databases offers improved data integrity and reduced redundancy while enabling institutions to monitor longer-term graduate success. Additionally, because they are based on a database, these systems allow for the generation of reports by institutions much quicker than traditional methods and also facilitate the maintenance of up-to-date alumni records over time.

Recent advancements of Web-based Information Systems have been incorporated with additional "Auto-Save" functionality to provide users with a better experience when completing surveys. Auto-Save functionality utilizes browser-based storage technologies and Database Synchronization methods to save the Users' partial responses as they complete their surveys. Web Usability studies demonstrate that Draft Auto-Save functionality can greatly decrease Survey Abandonment Rates and increase Completion Rates by providing protection from Data Loss due to Accidental Browsing Closures, Power Intermittency, or Unstable Internet connectivity. The Draft Auto-Save Functionality is especially helpful for Alumni Tracer System applications where Questionnaires are generally very lengthy.

Data analytics allows the institutional review of a much larger amount of alumni data, as well as the institutional use of this data to provide valuable insights from what were previously raw survey results. The institutional analysis through data analytics can determine an institution's employment patterns for graduates, track and assess graduate performance post-graduation, and also measure the success of each individual academic program. Through the application of various analytical techniques, organizations are able to identify both trends and relationships in very large datasets. Therefore, organizations will be able to make decisions based on evidence. Analytics provides support for

identifying areas of strength and weakness in the development of academic programs and preparation of graduates in the form of Alumni Tracer Studies.

One of the most widely utilized analytical methods for studying tracers is satisfaction distribution analysis. Satisfaction surveys often use Likert scales to measure graduate perceptions of curriculum quality, faculty performance, facilities, and institutional services. Mishra et al. (2020) have also mentioned that by using aggregation and descriptive statistics with satisfaction data, there are clear advantages in terms of learning about alumni's experiences and expectations. Satisfaction distributions provide insight into the various aspects of an institution that need to be improved and support quality assurance.

Graduate tracer surveys also serve as a means of measuring the success of institutions through graduate employment data. Typically, graduate tracer survey measures include employment rate, job relevance to the education program, income levels received by graduates, the type of industry or organization that hired graduates, and the time taken to find first employment after graduation. A key component of the Graduate Tracer Study framework developed by the Commission on Higher Education is to highlight the value of using employment-related metrics to determine if an institution's academic programs are adequately preparing its student body for future career opportunities.

To help facilitate understanding of analytical results and to make them accessible for stakeholders, many organizations have begun using Dashboard Technologies and Real-Time Reporting Systems. Dashboards are a way to visually display Key Performance Indicators (KPIs), such as statistics, trends, or charts/graphs. In their study on the use of real-time dashboards, Mahdizadeh et al. (2020) found that they assist with better decision-making based on being able to present updated data in a simple format. Administrators

may be able to review Response Rates, Employment Outcomes, and Satisfaction Levels through Visual Analytics without needing extensive technical knowledge.

Studies published recently show an increasing need for and application of Decision-Support-Systems (DSS), which support administrators' use of large amounts of data to interpret the data they are receiving from their students. DSS technology is a method used to help administrators generate recommendations for improvement within institutions using analytic output in conjunction with pre-defined rules and decision-making frameworks. The utilization of DSS generates more consistency and makes evidence-based decisions within institutions. Research has demonstrated that the use of decision-support tools improves the overall organizational effectiveness through lessening the subjectivity of decision-making and fostering data-driven planning processes.

Rule-based systems have been widely utilized as an approach to support decision-making within educational settings. Rule-based systems apply pre-defined If-Then logic to process data and provide users with automatic recommendations based on the data they input. Chen & Chang (2024) demonstrated how a rule-based recommendation engine can translate analytical insights into implementable actions through threshold-based criteria. As rule-based systems are transparent and interpretable, they are uniquely suited for educational environments in which the rationale supporting decision-making is critical for stakeholders.

Decision Support Systems with Recommendation Engines are a way to add to what DSSs do, as they provide decision makers with real-time information in response to the outcome of a survey. If survey responses reveal lower than established thresholds for overall satisfaction, a recommendation engine may then prompt decision-makers to implement additional curriculum offerings, develop faculty through training programs, make physical changes to facilities at an institution, or implement new methods of

supporting students. Chen and Chang (2024) demonstrated that using automated recommendation systems improves the speed at which organizations can respond to their own performance data because it shortens the amount of time needed to analyze survey data and create a plan for how to use the data to enhance an organization's performance.

Program improvement guidelines generated through rule-based decision-support systems have become increasingly valuable in quality assurance processes. Educational institutions can utilize alumni feedback to identify curriculum gaps, strengthen industry alignment, and enhance graduate employability. Zamora and Dorado (2015) emphasized that data-driven recommendations contribute significantly to continuous quality improvement by ensuring that institutional decisions are supported by empirical evidence rather than assumptions. Similarly, Cuadra et al. reported that systematic evaluation of alumni outcomes provides critical insights for strategic planning and academic program enhancement.

Taken together, the previous research on alumni tracing is important because it shows that the most successful alumni tracing programs are those where an alumni tracer program has all of the elements of a complete program, such as survey management, database management, data analysis, data visualization, decision support capabilities, and employment tracking. The majority of the research suggests that having a reliable database of alumni, good data visualization techniques, and ways to track what your graduates do after they graduate, along with ways to automatically recommend actions based upon the results of the data, will help institutions make informed decisions. This body of research provides substantial evidence that supports the need to develop a new alumni tracer study system called DOrSU Alumni Tracer Study System: A web-based graduate tracking and employment analysis platform.

2.3 Synthesis

Alumni Tracer Studies are becoming a more important resource for Academic Program Evaluation, Graduate Employability Measurement, and Institutional Planning within Higher Education. Alumni Tracer Systems provide colleges with key statistics about what their graduates are doing post-graduation (Employment Status, Career Advancement, Job Relevance, Graduates Satisfaction) and allow them to determine whether or not their academic programs are preparing students to meet the needs of the labor market. Many institutions, however, still face challenges when it comes to managing Alumni Data because they have traditionally used manual methods to collect data, keep records in different locations, and use paper-based survey instruments, which results in Alumni Data Sets that may be incomplete, inaccurate, and/or inconsistent, and therefore can limit an institution's ability to make decisions and analyze its data effectively.

The research emphasizes how institutional data collection can be enhanced using online multi-section surveys and a centralized relational database for collecting and storing data. The use of an online multi-section survey enables institutions to collect all graduate-level information from their graduates. A centralized relational database is used in order to provide a secure location where institutions can store all alumni data collected. This provides institutions with a better means of managing the data they collect, and increases both the data's accessibility and its ability to be retained over time. Auto-save capabilities will further increase the usability of online survey tools by providing users with some assurance that if they leave or exit a survey before completing it, their data will still be saved. Data collection and analysis are two primary areas of focus in the literature. Descriptive statistics, job placement rates, satisfaction ratings, and interactive dashboards represent just a few examples of analytical tools that can aid institutions in interpreting

their alumni data and, therefore, enable them to make quicker decisions based on data rather than instinct.

FURTHERMORE, the ability of institutions to transform data into action-relevant advice is enhanced through the incorporation of rule-based systems and decision support systems. The decision support systems are used by the analytical output of the data and by if-then rules in order to create automated suggestions for curriculum improvements, faculty professional development, and institutional development. The application of automatic suggestion engines will make it possible for responses to surveys to be faster and thus more responsive, as well as allow institutions to respond more quickly to the interpretation of their own survey results, thereby allowing them to make decisions with greater efficiency and based on evidence. Overall, the literature provides substantial support for developing a tracer system for alumni, which includes integrating all of the components of a tracer study, such as survey administration, centralized data storage, analytics, visualization, and decision support, creating a base upon which the proposed DOrSU Alumni Tracer Study System: A Web-Based Graduate Tracking and Employment Analysis Platform can be built.

2.4 Related System

Figure 3. Online Alumni Tracer System for Cavite State Graduates (JISEM, 2022)

Online Alumni Tracer System for Cavite State Graduates

The web-based Online Alumni Tracer System of Cavite State University - Imus campus is an internet application that supports the tracking of the current employment status of alumni of the university, in addition to assessing the degree of alignment of their present employment with their academic background. This provides the institution with timely and accurate data to assist in institutional planning. Through the use of agile development methodologies and standard web technology, this application automates the

process of collecting and analyzing data on alumni and significantly improves the accuracy of the data collected from alumni when compared to manual methods. Although the strength of the online tracer lies in its ability to automate data collection, track alumni employment status, and provide easy-to-use applications, there are no additional features available to support other functions of interest to alumni, including but not limited to alumni networking functionality, event management functionality, mentorship functionality, mobile app support, or secure verification mechanisms. In comparison to these limitations, the proposed DORSU Alumni Tracer has opportunities for improvement through the integration of real-time dashboard capabilities, communication tools, a secure registration mechanism, and engagement capabilities, providing an interactive, dependable, and data-driven system that will be able to monitor alumni careers.



Figure 4. PUP eAlumni Portal (Cavite State University, 2023)

PUP Alumni Portal

The eAlumni Portal offered by the Polytechnic University of the Philippines is an online registration system where alumni can register their names on the list of alumni; order Alumni ID Cards; get the Memorabilia Yearbook and Year Book of Graduates; obtain Documents such as Transcripts of Records and Diplomas; and produce Downloadable

Resumes through a website created specifically for the purpose of connecting alumni with service providers and providing them with information about events. While PUP's portal has the advantages of simplifying access to administrative services; enabling alumni to perform administrative functions themselves; and facilitating alumni association membership recruitment efforts; compared to other more complete alumni tracing systems (e.g., the one used by Cavite State University) it does not have the specialized capabilities such as Career Outcome Tracking, Employment Analytics, Field Alignment Assessment, Mentorship/Networking Module or Real-Time Dashboarding (Cavite State University, 2023; Polytechnic University of the Philippines, n.d.).

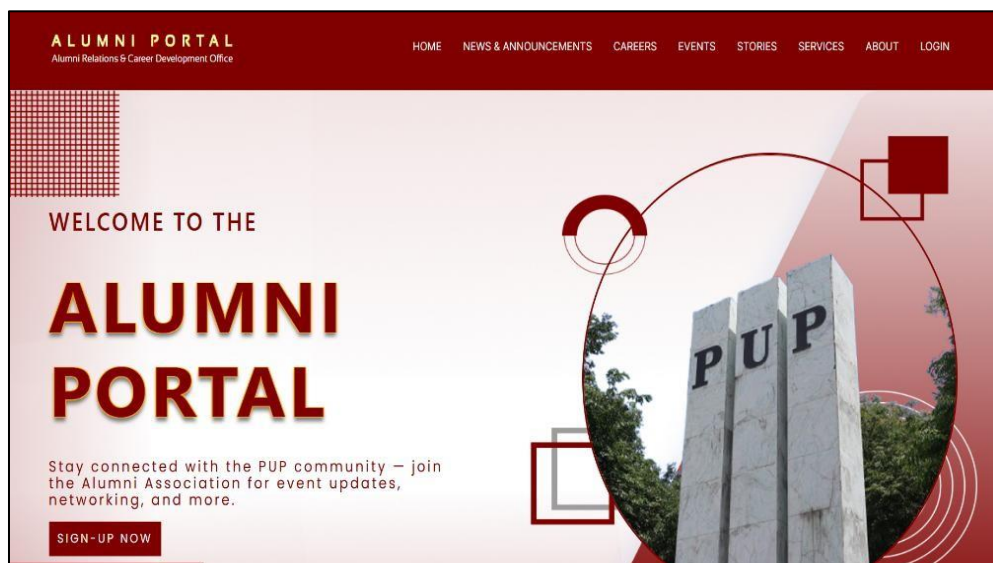


Figure 5. Alumni Tracking System for Aemilianum College Inc. (Beck et al., 2001)

Alumni Tracking System for Aemilianum College Inc.

The Aemilianum College Inc. (ACI) Alumni Tracer System (Arbaja, Sarmiento & Bausa, 2023) was an online program used to monitor graduates' employment status, their field of work, and where they are located using Visual Analytics. The system was created using Agile Development Methodology, which enables rapid and flexible development in

software projects (Beck et al., 2001). It has been successful in providing an accurate, reliable, and accessible means to track graduate placement and visualize data. However, while the ACI Alumni Tracer System provides a high level of automated tracking and data analysis capabilities, it does not provide a mechanism for engagement with alumni or tools for alumni networking, mentoring, creating events, or viewing real-time dashboards. The DOrSU Alumni Tracer System (proposed) integrates tracer-specific analytical functions with interactive modules designed to encourage engagement between alumni, students, and university staff. Additionally, the DOrSU Alumni Tracer will be designed to be responsive on all devices and include mechanisms to engage alumni actively and make the system a more dynamic and user-centered experience.

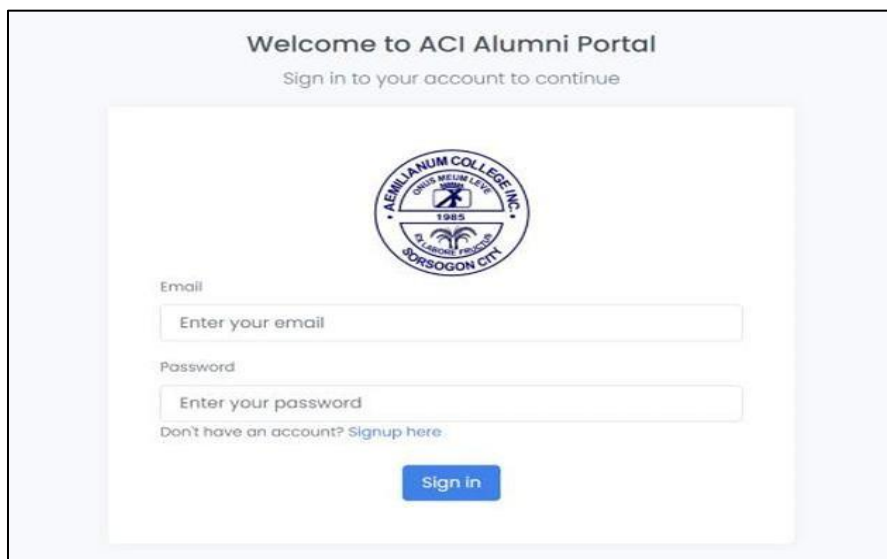


Figure 6. La Salle University–Ozamiz (LSU) eTracer (Villanueva, 2022)

La Salle University–Ozamiz (LSU) eTracer

The primary focus of the La Salle University-Ozamiz (LSU) eTracer is to conduct alumni research using surveys to identify alumni in their respective fields of study (engineering and graduate studies), employment status, career development, and job placement based on their degrees via an online form and/or social media (Villanueva,

2022). Although eTracer has been successful at collecting data for evaluating curricula and tracking the careers of graduates, it does not provide a central platform that includes automated dashboard reporting capabilities, mentoring functions, mobile access, or interactive opportunities for alumni engagement. As such, the DORSU Alumni Tracer System will fill this void by providing the tools needed to provide real-time reporting capabilities, secure student/alumni profile management, and interactive modules to create a dynamic connection between alumni tracking and career assessment.

Key Features	CvSU Alumni Tracer	PUP eAlumni Portal	ACI Alumni Tracer	LSU eTracer	DOrSU Alumni Tracer (Proposed)
Alumni Profile Management	Yes	Yes	Yes	Yes	Yes
Employment Data Tracking	Yes	Yes	Yes	Yes	Yes
Data Analytics & Visualization	Yes	No	No	Yes	Yes
Secure Data Validation	No	No	No	No	Yes
Decision Support System Integration	No	No	No	No	Yes
Feedback Collection	No	Yes	No	Yes	Yes
Mobile-Friendly Interface	Yes	Yes	No	Yes	Yes

Table 1. Comparison Table of Related Systems and the Proposed System

Table 1: The proposed DORSU Alumni Tracer System was compared to other Alumni Tracer Systems at CvSU, PUP, ACI, and LSU. Each of the systems can provide basic functions (alumni profiling; employment data), but each one has different levels of functionality.

Most of the Alumni Tracers were able to offer some or most of the same features (data analytics, feedback mechanisms, accessible through an app); none of them had integrated decision support, nor had they validated their alumni's data in a secure manner. It is on this basis that the current research has focused on providing alumni tracking and

analysis that would be driven by data to help institutions make informed decisions. The proposed DOrSU Alumni Tracer System was therefore intended to be able to track, analyze, and visualize alumni's career development to assist in making institutional planning and policy decisions.

CHAPTER III

METHODOLOGY

3.1 Software Methodology

In this research, the modified waterfall model provided a structured process with a sequential flow and enabled some degree of iterative refinement and revision across the various stages. The modifications to the original model gave the researchers the opportunity to return to earlier stages if issues arose, especially during the testing and validation phases. VERITAS 360 required flexibility in its methods, as the system used many different types of complex components, including automated evaluations, diagnostics, descriptive business analytics, and NLP-driven sentiment analysis. As stated by Steve McConnell (1994), software developed using a structured development process reduces risk, improves predictability of the project, and enhances the overall quality of the final product; thus, supporting the use of a highly regulated but flexible development method. Prior to moving from one stage to another, each stage was well documented and validated, so there could be no doubt about what was requested and how those requests were met. The loop-back aspects of the modified waterfall model allowed for corrections and improvements to occur in order to reduce errors and increase overall system quality.

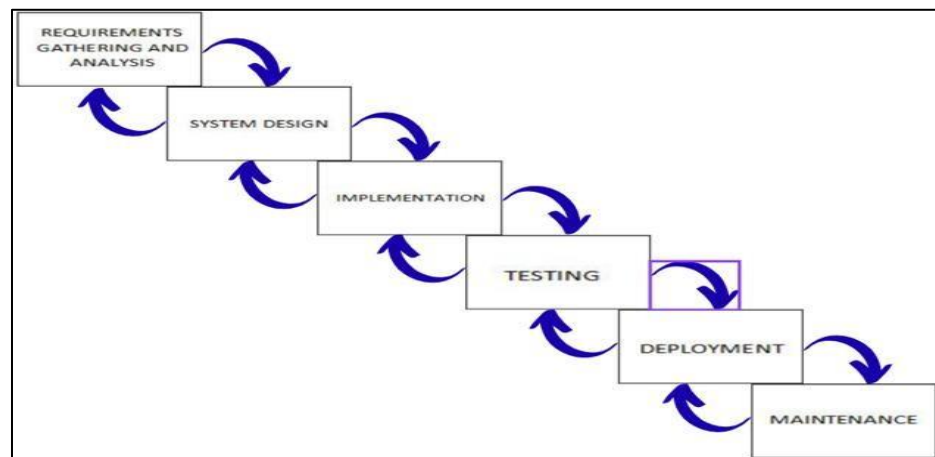


Figure 7. Modified Waterfall Model (McConnell, 1996)

This phase involved gathering detailed system requirements from university Alumni. These requirements include both functional (e.g., alumni profile submission, employment tracking, data visualization) and non-functional aspects (e.g., data privacy, usability, system availability). The outcome was documented in a Software Requirements Specification (SRS) document, which served as the foundation for the next development phases.

The design phase outlined the system's architecture, data flow, and interfaces. The DOrSU Alumni Tracer System used a three-tier architecture (frontend, backend, database) with React for the frontend, Node.js with Express.js for the backend, and MySQL Workbench for data storage. Design efforts also focused on responsive user interfaces, role-based dashboards (for alumni and administrators), and an interactive data analytics module to visualize employment trends.

The System Development Team designed and built the system in accordance with the design specifications during the implementation of the project. The team used the chosen technology solutions to develop the system. For example, Microsoft's Visual Studio Code was the primary integrated development environment (IDE) that provided the necessary tools for code editing, debugging, and source control (Microsoft, 2024). Additionally, GitHub was utilized to collaborate on source control and implement continuous integration. The team also developed React components to create a user interface that was consistent and responsive. Lastly, the team created backend APIs utilizing Express.js and Node.js to manage the business logic and communicate with MySQL Workbench (Sharma & Kumar, 2022). The team followed modular coding practices along with best practices for maintaining and testing the system (Osmani, 2020).

Comprehensive testing was conducted to ensure system correctness, reliability, and performance. Unit testing, integration testing, and system testing were performed using

modern testing tools and frameworks. Testing focused on verifying that all features, such as request submission, document generation, notification, and reporting, functioned as expected, and that security and privacy requirements were met (Pressman & Maxim, 2020). User acceptance testing (UAT) was conducted with representatives from university offices and ICTU personnel to validate the system in realistic use cases (Sommerville, 2016).

After successful testing and receiving the necessary approval from stakeholders, the DORSU Alumni Tracer System was deployed to production. It was also developed with strong monitoring capabilities and several backup options. In addition to maintaining logs for incident tracking, the system continuously monitored itself for problems. System maintenance consisted of implementing bug fixes, enhancing features, and updating the system as needed to comply with new ISSP policies and user needs. Because the modified waterfall model is an iterative process, it allows for the systematic planning and implementation of future enhancements.

3.2 System Planning

The planning of this stage laid the foundation for the entire project. It involved defining the project's concept, conducting the feasibility study, and developing the key management tools such as the Work Breakdown Structure (WBS) and the Gantt chart.

Project Conceptualization

The concept of developing a DORSU Alumni Tracer System emerged upon recognizing the inefficiencies in the other universities' existing system processes involving the tracing, monitoring, job matching, and collecting surveys of how well our DORSU Alumni have been doing as of today. The growing number of DORSU Alumni Graduates and the increasing administrative workload necessitated a shift toward a centralized and automated system for monitoring.

By monitoring the tasks of the alumni and observing the current manual workflow, it became clear that a digital platform is essential for the system. The goal is to move away from manual encoding and instead provide a web-based platform where every alumni event can be posted and shared. This new platform will offer real-time survey monitoring and dashboards for both Program Heads and Administrators. This dashboard allows them to track employment outcomes such as the duration it takes for graduates to find their first job, their monthly salaries, and their current professional levels. Beyond just tracking, the system will generate reports, improve transparency, and make services more accessible.

Feasibility Study

The feasibility of the study assesses whether the project can be realistically developed and implemented by examining the operational, economic, and technical constraints.

Operational Feasibility

The DORSU Alumni Tracer is an important tool for the University, mainly because the administrators are the ones who would be using it. Today, as part of its current processes, the university has to manually manage its alumni data. For example, we have to spend time writing out and collecting paper-based surveys, sift through numerous links on the Internet, and enter the same information repeatedly in old Excel files. With this new automated process, our staff will no longer be able to focus on these tedious tasks. Rather than spending countless hours on the telephone or composing hundreds of e-mails, which ultimately results in tracking a graduate's information, the staff will simply allow the system to do all of that. While it may seem simple, having fewer tedious tasks does make their job easier, but most importantly, it will result in much better accuracy with regard to the data collected; plus, it provides a single location where all of this information is stored.

Organizational Support

The university administrator demonstrated strong organizational support for implementing the alumni tracer system and also provided a collection of alumni graduates' datasets. Administrators recognize its role in enhancing data management and improving alumni engagement of the Davao Oriental State University (DORSU).

Economic Feasibility

A Cost–Benefit Analysis (CBA) using default academic estimates was conducted to assess the system's financial viability.

Table 2. Cost–Benefit Analysis (CBA)

COST	
Category	Estimated Budget
1. Requirements Gathering and Analysis	—
2. System Design	
3. Implementation	—
3.1 Web Hosting Annual Subscription	Php 10,0S00.00
4. Testing	—
5. Deployment	—
5.1 Conduct of User's Training	Php 3,000.00
5.2 User Evaluation	Php 1,500.00
6. Maintenance	—
6.1 Development & Distribution of User Manual	Php 2,000.00
6.2 Annual Maintenance and Updates	Php 25,000.00
Total Cost:	Php 41,500.00

BENEFITS	
Time Savings Calculation:	
Manual process: 10 hours	
Automated system: 2 hours	
Savings per month: 8 hours	
Salary basis: Php 30,705 / 22 working days $\approx$ Php 1,395.68 per hour-equivalent	
Monthly Savings = $8 \times 1,395.68 =$ Php 11,165.45	
Annual Savings:	
$11,165.45 \times 12 =$ Php 133,985.40	
Return on Investment (ROI):	
$ROI = \frac{133,985.40 - 41,500}{41,500}$	
Payback Period:	
$Payback\ Period = \frac{41,500}{11,165.45}$	
$= 3.72\ months$	

Table 2 presents the cost–benefit analysis of the proposed DORSU Alumni Tracer System, highlighting its financial feasibility and operational value. The total cost of developing and deploying the system is Php 41,500.00, which includes the annual web hosting subscription, the database subscription, user training, evaluation, production of user manuals, and one year of system maintenance and updates.

Financial Return on Investment (ROI) The operational costs associated with the time spent on the above-mentioned processes will be greatly reduced through the application of the proposed system. Through this process, it is estimated that all of the previously stated tasks can be completed in two hours, thereby reducing the total amount of time expended by one hour each month. For example, it is estimated that there would be a value of PHP 11,165.45 saved per month as a result of this time saving. It is further estimated that the total value of the time saved would be PHP 133,985.40 annually.

Therefore, based upon this estimate, the proposed DORSU Alumni Tracer System has provided an estimated return on investment of 133.1%. Thus, the system has provided an extremely high rate of financial return on investment in the first year after its implementation. In addition to the high rate of financial return on investment, the payback period of approximately three and seventy-two months (3.72 months) indicates that the cost of implementing the proposed system was recovered in less than four months after its implementation. These results indicate that the DORSU Alumni Tracer System provides many additional operational advantages to the University.

Technical Feasibility

The project is technically feasible because the tools and technologies required are available, reliable, and compatible with the researchers' skills.

Hardware Compatibility

- Standard desktop/laptop and mobile
- Server hosting environment

Table 3. Technical Feasibility

Tool	Purpose
Django (Python)	Backend development
MySQL Workbench	Database management
React	Frontend
VS Code	Integrated development environment

Table 3 presents the technical feasibility of the proposed DORSU Alumni Tracer System by identifying the key development tools and their respective functions in the system architecture. Django, a Python-based web framework, is used for backend

development as it supports secure, scalable, and efficient handling of server-side logic, APIs, and user authentication processes. MySQL Workbench is utilized for database management, allowing developers to design, organize, and maintain structured alumni data efficiently while ensuring data integrity and reliability.

A combination of two technologies is used to develop the User Interface: Visual Studio Code (the IDE) and React (a Frontend Framework). These allow for a fast, friendly, user-responsive interface between users and the system. Together, they provide an overall solution that supports all of the technical, modern, and functional features that will be required by the Alumni Tracer System.

Capability of Tools

- Django provides a robust framework for building secure server-side applications.
- Using a Django (Python), Rule-based algorithm implemented using conditional logic in Python
- MySQL Workbench is a visual database design and administration tool that is used for creating, managing, and optimizing MySQL databases.
- React is one of the best tools for making responsive and component-based user interfaces.
- VS Code is a lightweight and yet VS Code is a powerful code editor that supports multiple programming languages and frameworks.

Work Breakdown Structure (WBS)

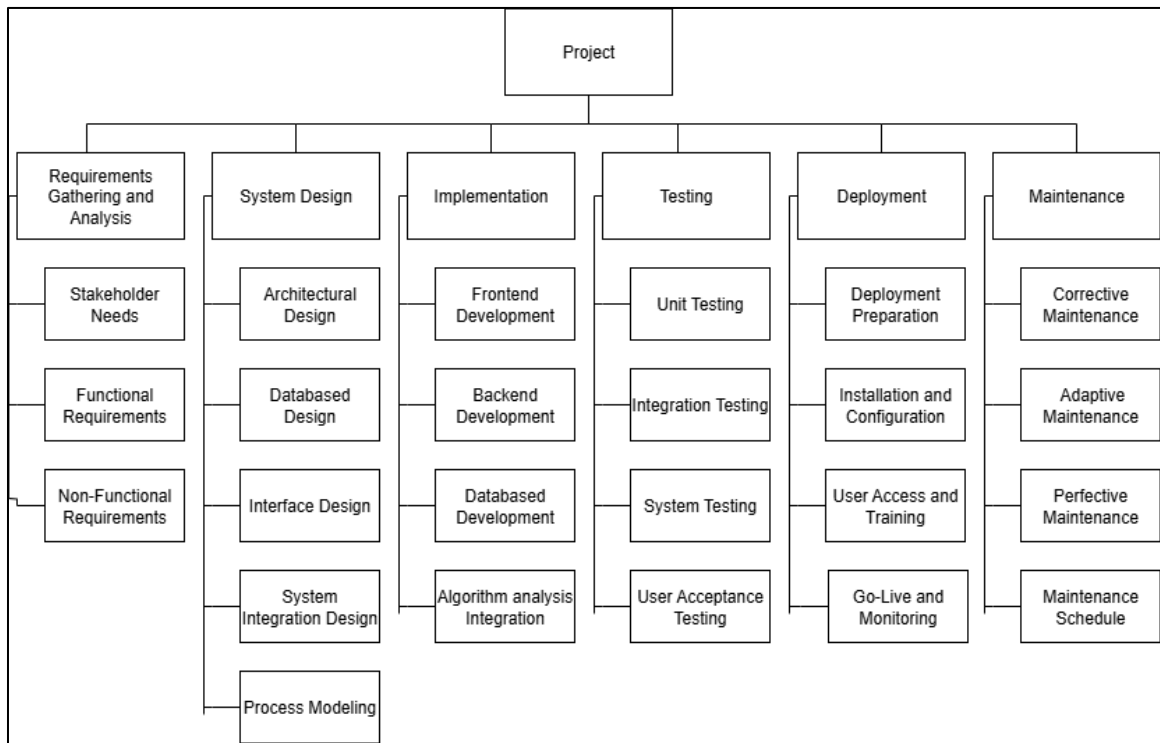


Figure 8. Work Breakdown Structure (WBS)

Figure 8 illustrates the System Development Life Cycle (SDLC) of the DORSU Alumni Tracer System, showing the sequential phases involved in its development. It begins with Requirements Gathering and Analysis, where stakeholder needs and both functional and non-functional requirements are identified. This is followed by System Design, which includes architectural, database, interface, system integration, and process modeling to define how the system will function.

The next phase is Implementation, where the system is developed through frontend, backend, and database coding, along with algorithm integration. This is followed by Testing, which ensured system quality through unit testing, integration testing, system testing, and user acceptance testing. After testing, the system proceeds to Deployment, where it is installed, configured, and prepared for user training and go-live monitoring. The

Maintenance phase ensures continuous improvement through corrective, adaptive, and perfective maintenance, supported by a scheduled maintenance plan to sustain system performance and reliability.

Gantt Chart

The Gantt chart in Figure 9 presents the **System Development Timeline (Gantt Chart)** of the DORSU Alumni Tracer System, showing the schedule of activities from **August 2025 to June 2026**. It illustrates the project's step-by-step progression across development phases, with each task aligned to a specific month to ensure an organized implementation.



Figure 9. GANTT CHART

3.3 System Analysis

The system focused on understanding the university's existing process for tracking graduates and collecting alumni information. This shows how the institution gathered, stored, and used alumni records, as well as the challenges encountered in monitoring graduate outcomes. Information was collected through the university-provided survey forms. The findings from these forms served as the basis for identifying the requirements of the proposed Alumni Tracer System and addressing the limitations of the current alumni tracking process.

Description of the Current System

The current alumni system at the university is primarily manual, and it only relies on forms, spreadsheets, and unstructured electronic records. Graduate information is initially collected during graduation and stored in spreadsheets and printed documents. To gather employment and career-related information, alumni coordinators manually contact graduates through social media platforms, email, phone calls, and Google Form surveys.

BPMN of Current System (Narrative)

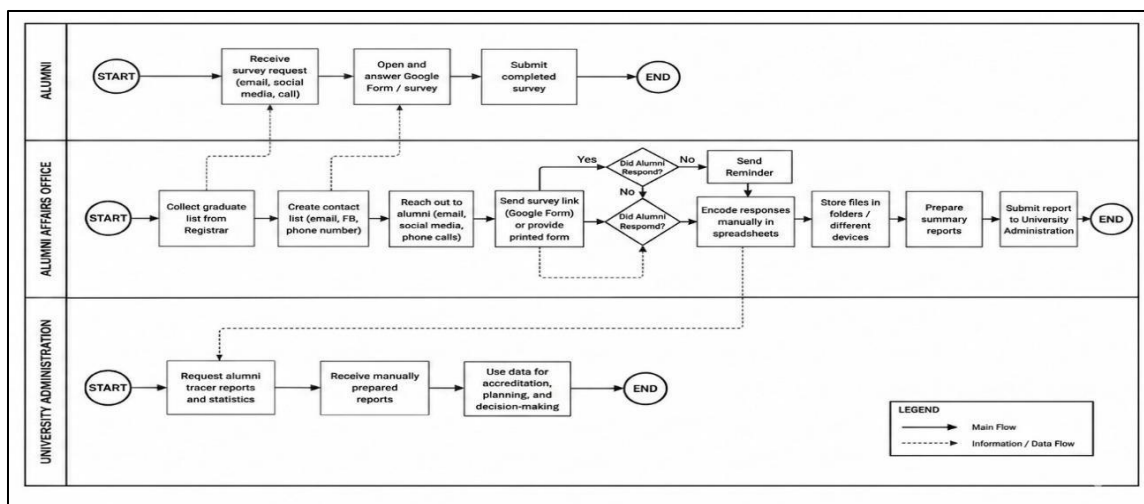


Figure 10. BPMN of the Current Alumni Tracing Process

The Graduate Tracking Process within the university has been mapped as it exists today (see figure 10) to illustrate the processes followed by Alumni Affairs Office personnel, Alumni Affairs Office Staff, and University Administration. It illustrates how surveys are distributed from Alumni Affairs, how responses are collected or tracked using the Reminder Loop to track when follow-up communications were sent, and how this manual response collection was encoded, stored, and compiled into reports. This illustration visually demonstrates inefficiencies in the existing system and supports our argument for a centralized, analytics-driven approach to manage all data related to the tracking process.

Requirements Specification

The Requirements Specification acts as the blueprint for the DOrSU Alumni Tracer; this is what the system needs to do and how it should perform to be effective. By separating these into functional features and technical standards, we provide a structured guide to ensure that the platform is not only helpful for users but also a reliable foundation.

A. Functional Requirements

The system shall allow:

Alumni

- Register and log in
- View post
- Answer digital surveys
- Change password and profile image.
- View academic page.

- View About us.
- View and can submit a contact

Admin

- Create, list, read, update, and delete alumni and program-head users
- Review and change approval
- Change password and profile image
- Create, edit, and delete posts (announcements) with images
- View dashboard aggregates and manage users.
- Delete user records

Program Head

- View and update their own profile
- Change password and profile image.
- See assigned programs and their status.
- See program status (pending, approved, rejected).
- View alumni lists and details for their assigned programs.
- Review alumni survey submissions
- View basic aggregates (number of alumni, survey results) for their assigned programs.

B. Non-functional Requirements

- **Performance:**
 - Fast response times for dashboard, user management, and survey aggregation.
 - Efficient database queries (uses Django, aggregation, and pagination).
- **Scalability:**
 - Supports growth in alumni, program heads, and survey data.
 - Can be deployed on scalable hosting (PythonAnywhere).
- **Security:**
 - Role-based access (admin, program head, alumni).
 - Data privacy for alumni surveys and user info.
- **Reliability:**
 - Handles errors gracefully (validation, DB errors, network issues).
 - Robust fallback logic for legacy tables and API endpoints.
- **Usability:**
 - Intuitive dashboard UI for admins, program heads, and alumni.
 - Responsive design for desktop and mobile.
- **Maintainability:**
 - Modular codebase (separate backend/frontend, clear API endpoints).
 - Uses Django and React best practices.
- **Portability:**
 - Can run on Windows, Linux, or cloud platforms.
 - Database-agnostic (MySQL).
- **Auditability:**
 - Tracks user actions (survey submissions, status changes).

- Centralized notifications and logs.
- **Availability:**
 - Designed for high uptime; can be monitored and restarted easily.

Proposed Solution

Deliver a secure, maintainable, and deployable Alumni-Management web system built on the existing Django backend and React frontend. The solution keeps the current data model and UI where possible, hardens authentication/authorization, improves robustness around legacy-table handling, and prepares the app for production deployment (Hostinger or similar). The approach is quick and low-risk fixes first (security + token enforcement), then the operational setup (static files, DB), and the polish (monitoring, tests, exports).

3.4 System Design

The system uses a python Django rest framework for backend APIs and react for the frontend, supporting the role-based access for admins, alumni, and program heads. Data is stored in MySQL Workbench, with token authentication for security. Admins manage users, posts, and surveys; alumni interact with posts and profiles. The architecture is modular, with clear separation of backend and frontend, and is ready for deployment on Hostinger.

System Architecture

Figure 9 illustrates the system architecture, a modular client-server design. The system has three main types of users. The admins, who manage accounts for the users and send out alerts. The alumni, who fill out surveys and update their profiles, and program Heads, who look at special reports to see how graduates are doing in their careers. Users

log in using a regular web browser like Chrome or Safari, and all data is sent over a secure Internet (HTTPS) connection to keep it safe.

Once the information is sent, it goes to a Web Server that uses a Django REST API to organize the data. All of this information, like user details and survey answers, is kept in a MySQL Database. To make the system smart, it uses a Rule-based Algorithm to look at things like a course taken and a current job to predict where they are matched. Finally, an Analytics Survey turns these complex numbers into simple charts and reports so that school leaders can make better decisions for their students.

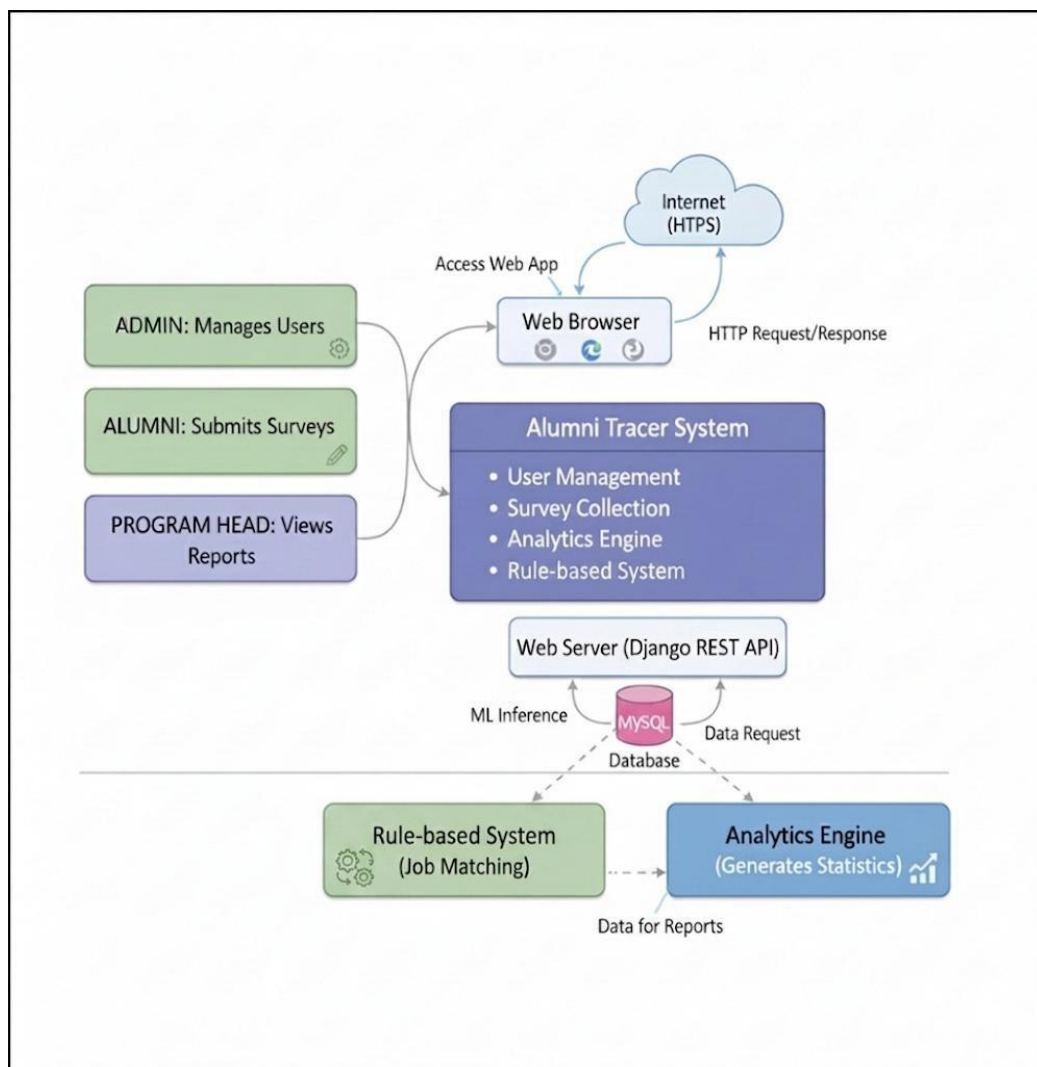


Figure 11. System Architecture

Users' alumni table - contains alumni-specific data (personal details, graduation year/program, contact information, employment status). Each user account has one record in this table; each record in this table will be related to an authenticated user.

Users_employmentrecord table - contains records relating to a graduate's employment experience, including company name, position held at that time, employment status at the time, salary range, and date of hire. The two tables have a relationship, so the graduate can be tracked from the point they graduated until the current day regarding their employment experience.

The users_alumnisurvey is a database table where all of the alumni surveys are stored. In this table, you will find demographic, educational, and employment-related information that has been provided to us by alumni. Alumni survey information can be used to create reports, generate analytics, and assist in making decisions within the institution. There are several additional tables in the database that support authentication and authorization. Some of these include; auth_permission, auth_group, auth_group_permissions, auth_user_groups, and auth_user_user_permissions. All of these tables serve to assign permission levels to each user role, which then limits what each user may do on the system.

The django_admin_log table is a record of all things that have been done by admins in the system, as well as an audit trail (a history) of all changes made to the system. The authtoken_token table keeps track of authentication tokens, which are used for safe login sessions and other forms of access via APIs. In addition to its administrative capabilities, the system has additional features that provide support for community interaction with the use of posts_post, posts_comment, and posts_like tables. These tables will enable users to make posts, comment on discussions (posts), and like or dislike posts, allowing alumni and administrators to communicate amongst themselves.

Data Dictionary

The data dictionary provides a detailed description of the tables, fields, and data elements used in the database. In this study, the data dictionary was used to define all database components of the Alumni Tracer System, including alumni records, employment information, user accounts, survey responses, and other relevant data. This documentation supports efficient database maintenance, system development, troubleshooting, and future enhancements.

Admin Table

Table 4. Admin

Attribute	Content/Description	Type	Format	PK	FK
id	Admin ID (auto)	Integer	AutoField	Yes	
username	Admin username	String	max_length=150		
email	Admin email	Email	unique		
password	Admin password	String	max_length=128		
full_name	Full name	String	max_length=255		
image	Profile image (optional)	ImageField	upload_to		

This database is used to manage, validate, and securely store the core authentication credentials, identity information, and profile assets for the system administrators. It serves as the central security and identity repository for administrative access control.

Alumni Table

Table 5. Alumni

Attribute	Content/Description	Type	Format	PK	FK
id	Alumni ID (auto)	Integer	AutoField	Yes	
username	Alumni username	String	max_length=150		
email	Alumni email	Email	unique		
password	Alumni password	String	max_length=128		
full_name	Full name	String	max_length=255		
program_course	Program/course	String	max_length=255		
image	Profile image (optional)	ImageField	upload_to		
year_graduated	Year graduated (optional)	SmallInteger	nullable		

This database entity is designed to store and manage important information about alumni, including their login credentials, personal details, academic records, and profile pictures. It helps ensure that the alumni information is organized and secure while providing authorized access to the system. As the main repository of alumni data, it plays a vital role by maintaining alumni profiles and supporting the overall functionality of the Alumni Tracer System.

Program Head Table

Table 6. Program Head

Attribute	Content/Description	Type	Format	PK	FK
id	ProgramHead ID (auto)	Integer	AutoField	Yes	
username	ProgramHead username	String	max_length=150		
name	First name	String	max_length=255		
surname	Surname	String	max_length=255		
mi	Middle initial (optional)	String	max_length=1		
gender	Gender	String	max_length=10		
contact	Contact number	String	max_length=20		
email	Email	Email	unique		
faculty	Faculty	String	max_length=100		
program	Program	String	max_length=100		
image	Profile image (optional)	ImageField	upload_to		
status	Approval status	String	max_length=20		

This database entity will allow us to track all of the user's information regarding the Program Head. The database will contain the account username/password as well as other program head-related information, such as address/contact info/department. This will help ensure we have an accurate, organized, and secure source of program heads' information. Additionally, this will also provide the ability for the system to grant each program head the proper level of access and rights within the application. We can then use these records to effectively maintain the academic leadership accounts as well as assist with the responsibilities of managing Alumni Data and Activities.

Alumni Survey Form Table

Table 7. Alumni Survey

Attribute	Type	Format / Constraints	PK	FK (references)
id	AutoField (Integer)	Auto-increment primary key	Yes	No
alumni	ForeignKey	Nullable; SET NULL on delete; related_name=surveys	No	<u>users_alumni(id)</u>
last_name	CharField	max_length=255	No	No
first_name	CharField	max_length=255	No	No
middle_name	CharField	max_length=255; blank allowed	No	No
year_graduated	CharField	max_length=8; blank allowed (string/year)	No	No
course_program	CharField	max_length=255; blank allowed	No	No

Table: Comment

Attribute	Content/Description	Type	Format	PK	FK
id	Comment ID (auto)	Integer	AutoField	Yes	
post	Related post	ForeignKey			Post(id)
author_name	Author name (optional)	String	max_length=255		
author_id	Author ID (optional)	Integer			
text	Comment text	Text			
created_at	Creation timestamp	DateTime	auto_now_add		

Table: Post

Attribute	Content/Description	Type	Format	PK	FK
id	Post ID (auto)	Integer	AutoField	Yes	
title	Post title	String	max_length=255		
content	Post content	Text			
created_at	Creation timestamp	DateTime	auto_now_add		

Table: Like

Attribute	Content/Description	Type	Format	PK	FK
id	Like ID (auto)	Integer	AutoField	Yes	
post	Related post	ForeignKey			Post(id)
user_id	User ID (optional)	Integer			
created_at	Creation timestamp	DateTime	auto_now_add		

Table: PostImage

Attribute	Content/Description	Type	Format	PK	FK
id	Image ID (auto)	Integer	AutoField	Yes	
post	Related post	ForeignKey			Post(id)
image	Image file	ImageField	upload_to		

This entity is designed to store and manage the information collected from alumni through the tracer survey. It records the responses submitted by each alumnus together with the academic profile details, ensuring that the data is accurate and properly organized. By linking survey responses to the registered alumni records, the system can effectively track graduate outcomes and generate reports that support institutional planning and decision-making.

Data Flow Diagram (DFD)

- **Level 0 DFD (Context Diagram)**

Figure 13 shows the Alumni Tracer System as a single process interacting with the external environment.

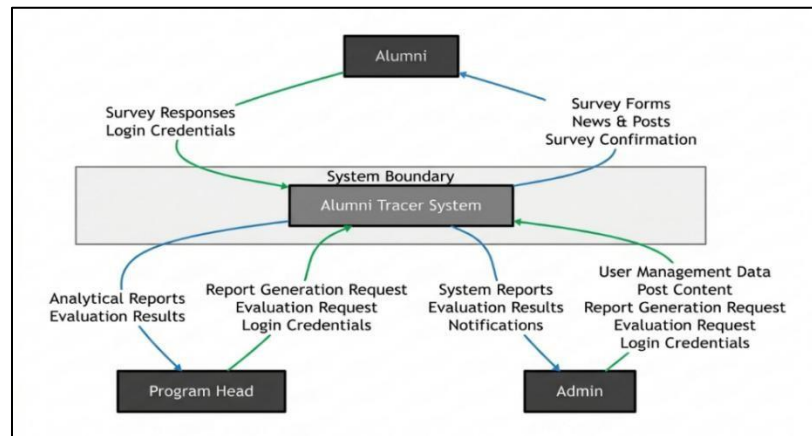


Figure 13. Level 0 DFD (Context Diagram)

- **Level 1 DFD**

Decomposes the system: Manage Alumni Information, Survey Management, Analyze Job Matching, and Generate Reports.

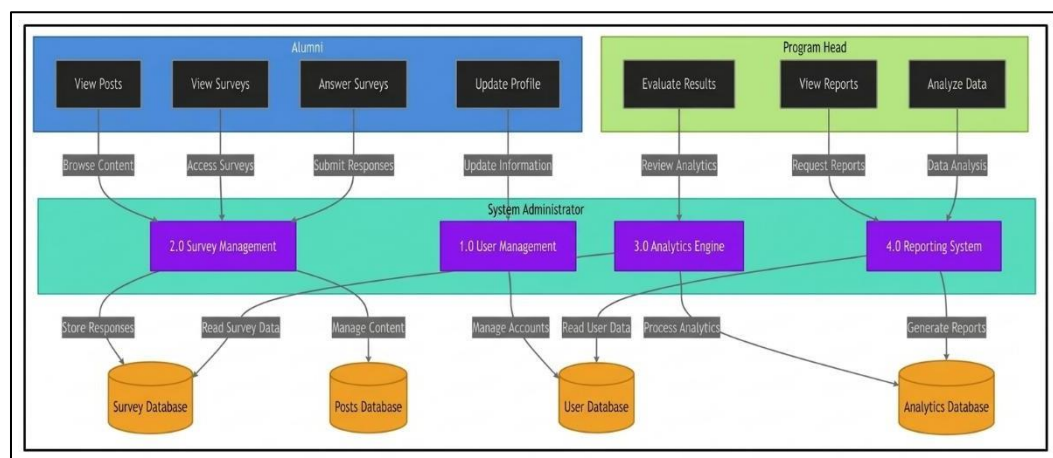


Figure 14. Level 1 DFD

Use Case Diagram (Narrative)

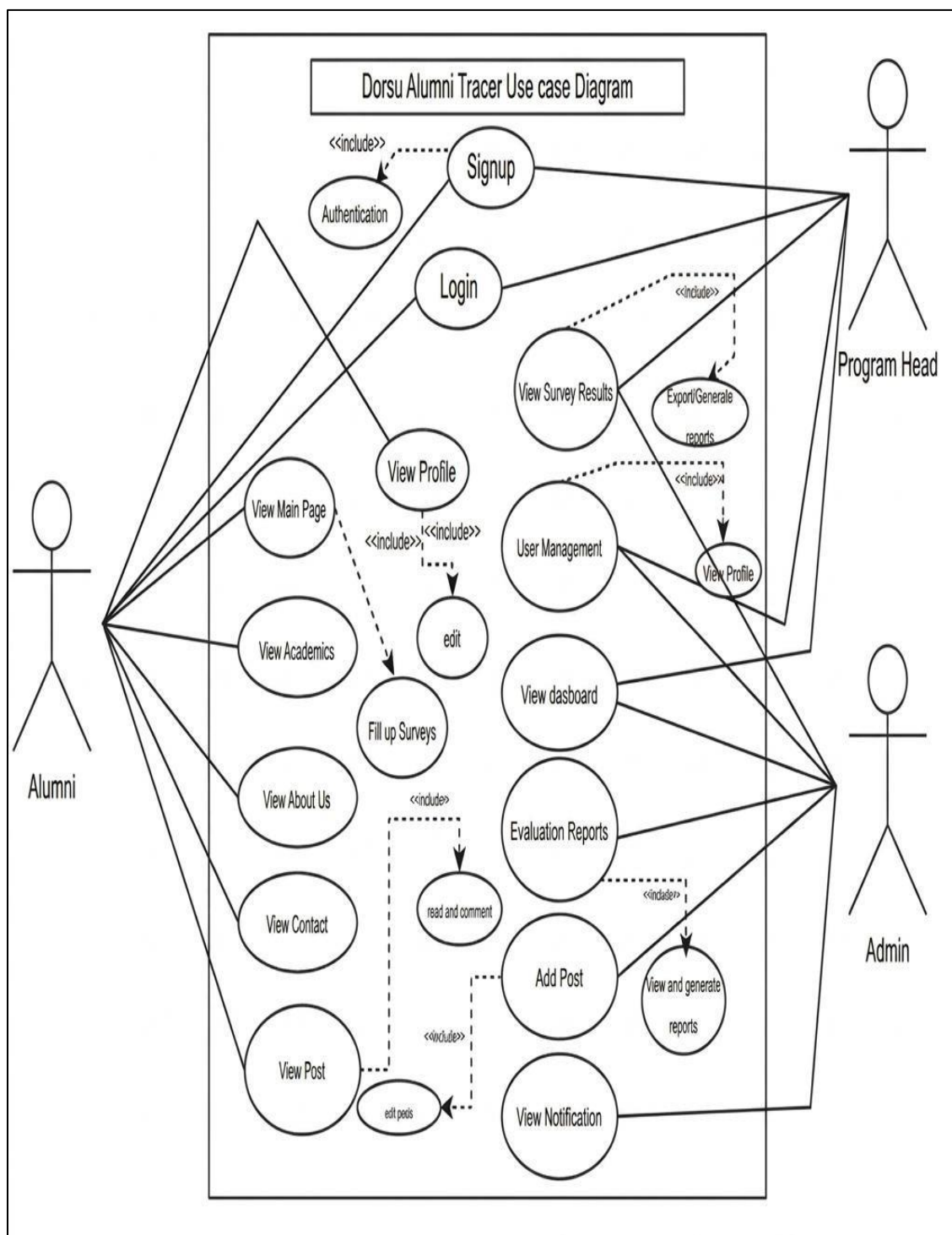


Figure 15. Use Case Diagram (Narrative)

Figure 15 illustrates that alumni are the main contributors. After a simple Signup process, they can log in to manage their own Profiles. Their most important task is filling out the surveys. They can also stay connected with the community by viewing, reacting to, or commenting on Posts. The admin is the person behind the scenes, making sure everything runs smoothly. They handle User Management and post updates for everyone to see. To see how the system is doing, they use a Dashboard and look at Evaluation Reports, which help them create detailed summaries of all the data being collected. The Program Head focuses on the User Management of the users, similar to the Admin. They also use the system to check Survey Results and the Dashboard. When they need to present this data, they can also export and generate reports, which gives them a clear, organized look at the alumni's progress.

UX Sitemap

Figure 16 outlines the hierarchical structure of the application's user interface, starting from the Login/Register page.

The sitemap serves as a guide for the application's layout and organizes the user interface into a clear structure that begins at the Login and Registration page. From this starting point, it maps out the path a user takes, showing how different sections of the platform connect and ensuring that navigating through the various features is organized.

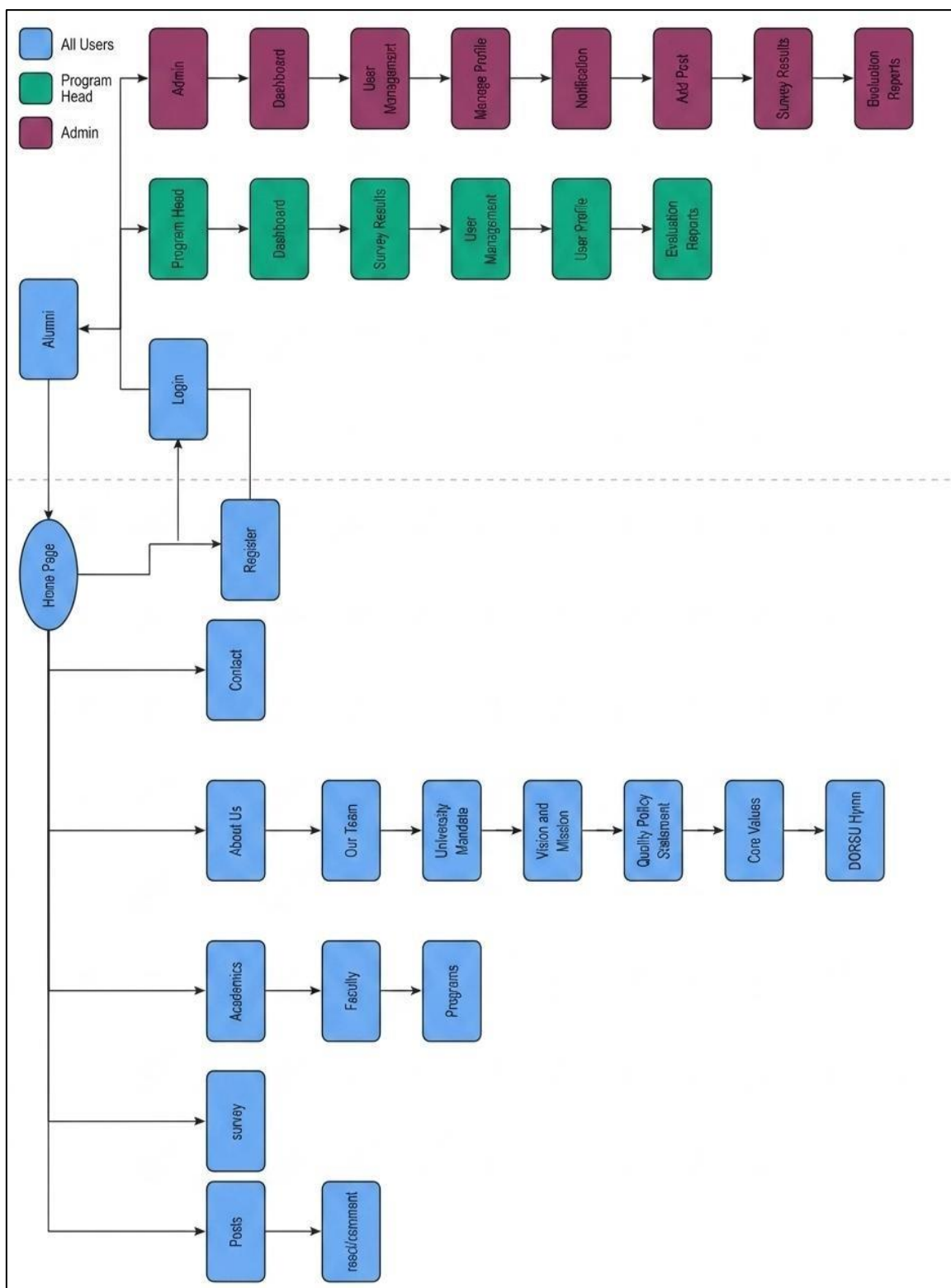


Figure 16. UX Sitemap

3.5 System Implementation

This section focuses on the actual development, testing, and deployment of the system.

System Development

The system was developed using modern web technology. The Frontend was built with React to create a responsive and consistent user interface. Backend was implemented using the Django framework in Python, which handled routing, session management, and business logic, and the integration of the Rule-based Algorithm for job matching and recommendations. The MySQL relational database was used to store all structured data, ensuring data integrity and efficient queries. including user profiles, job listings, and matching, was persisted in a MySQL relational database. Development was conducted in the Visual Studio Code IDE.

System Testing

To ensure the Alumni Tracer System is both reliable and user-friendly, researchers followed a thorough testing plan to verify that it works exactly as intended.

Technical Testing

The technical testing was broken down into three main stages to cover every corner of the system's functionality:

First, Unit Testing was conducted to check the individual blocks of the platform. The login and authentication modules were tested to ensure that the accounts remain secure, and verified that the core record management functions, such as adding, editing, or deleting alumni information, are working properly. We also checked that education and

employment tracking accurately saved data and that report generation functions worked as expected. By testing, it is ensured that each feature is strong and free from error before connecting it to the rest of the system.

Next, integration testing was conducted to make sure all the different parts worked together. During this stage, it was verified that the information flowed across the system. For example, it was ensured that data entered into an alumni registration form was correctly stored in the database and that the report modules could retrieve that information with correct accuracy. The connection between the React frontend and the Django backend was also tested, ensuring that both communicated effectively through APIs so that the user experience remained smooth and responsive.

Finally, system testing was conducted so that the platform was completely walked through from end to end. All different scenarios for the site were tested, such as logging in and managing profiles, monitoring career paths, and using filters to search specific alumni; also, the ability of the system to generate reports needed by institutional planners and future analysis was confirmed. This final check demonstrated that the system met all its functional standards and was ready for daily use by the university.

B. User Testing and Evaluation

```
=====
TESTING JOB MATCHING SERVICE
=====

Testing job matching on survey 1...

Label: match
Score: 66/100
Matched keywords: 1 found
Top 3 suggested roles:
  1. Academic Support / Tutoring Services
  2. Assessment & Research in Education
  3. Classroom Teaching

Reason: Matched: "Teacher" is listed under possible jobs for Bachelor of Elementary Education
      BEED). (Similarity: 66%)
```

Figure 17. Unit Testing Using Jest

```

===== test session starts =====
platform win32 -- Python 3.13.3, pytest-9.1.0, pluggy-1.6.0 -- c:\Users\kei\AppData\Local\Programs\Python\Python313\python.exe
cachedir: .pytest_cache
django: version: 6.0.1, settings: alumni_backend.settings (from ini)
rootdir: C:\capstone 2 Alumni\backend
configfile: pytest.ini
plugins: anyio-4.8.0, cov-7.1.0, django-4.12.0
collected 10 items

users/test_pytest_demo.py::TestProgramHeadAPI::test_list_program_heads PASSED [ 10%]
users/test_pytest_demo.py::TestProgramHeadAPI::test_create_program_head PASSED [ 20%]
users/test_pytest_demo.py::TestSurveyResults::test_survey_count PASSED [ 30%]
users/test_pytest_demo.py::TestSurveyResults::test_employment_calculation PASSED [ 40%]
users/test_pytest_demo.py::TestAuthentication::test_unauthenticated_access PASSED [ 60%]
users/test_pytest_demo.py::TestAuthentication::test_authenticated_access PASSED [ 70%]
users/test_pytest_demo.py::test_satisfaction_threshold[1-True] PASSED [ 80%]
users/test_pytest_demo.py::test_satisfaction_threshold[3-True] PASSED [ 90%]
users/test_pytest_demo.py::test_satisfaction_threshold[4-False] PASSED [ 95%]
users/test_pytest_demo.py::test_satisfaction_threshold[5-False] PASSED [100%]

```

Figure 18. Unit Testing Using Jest

User Acceptance Testing (UAT) was conducted for the university administrative staff. These participants were chosen for direct involvement in alumni data management.

A questionnaire aligned with the ISO/IEC 25010 Software Quality Standard was developed and validated by IT experts. The instrument measured system acceptability across (Functionality, Reliability, Usability, Performance Efficiency, and Security).

A 5-point Likert Scale (1-Strongly Disagree to 5-Strongly Agree) was used to assess user satisfaction. Data collected from the evaluation were analyzed using mean scores and descriptive interpretation to determine the overall acceptability of the system.

System Deployment

The Alumni Tracer System was deployed using the PythonAnywhere cloud platform to ensure that the application was accessible and reliable for all users. The

deployment began with the setup inside the server environment, where the PythonAnywhere virtual environment was configured, including Python and the Django framework, alongside the frontend integration of Node.js and React.

Once the environment was ready, database configuration followed by setting up and securing a MySQL within the platform to manage the system's data. During the deployment, the backend and frontend code were uploaded to the server, and the API connections, environment variables, and WSGI configuration were finalized to make the site live.

After ensuring that the deployment step was successful and smooth, a User Orientation phase was conducted through training to help everyone get comfortable with the system. This was supported by documentation, including a User Manual, an Administrator Manual, and a Program Head Manual.

3.6 Maintenance

To keep the Alumni Tracer System running, a simple and effective maintenance plan will be in place. This starts with the Error Handling, where a clear process is established for the users to report any bugs or glitches they find. We will also perform system updates to add fresh features, update any security, and make the system faster based on user feedback.

Finally, a Sustainability Plan will ensure the system's future by formally handing over management duties to the university's IT or management team, making sure it remains a helpful tool for years to come.

CHAPTER IV

RESULTS AND DISCUSSIONS

4.1 Design and Development of the System

Build a secure web-based system

Figure 17 shows the Create Alumni Account Module, which enables users to register and access alumni services. It provided fields for personal details, account credentials, academic history, and account security. The module ensured that every registering alumnus was properly linked to their respective program and graduation year, strengthening data integrity.

Create Your Alumni Account
Fill out the form below to register and access alumni services.

PERSONAL INFORMATION
First Name *
Last Name *
Middle Initial

ACCOUNT INFORMATION
Username *
Email Address *
user@example.com

ACADEMIC INFORMATION
Program / Course *
Search or select your program
Year Graduated *
Select year

SET PASSWORD
Password *
Minimum 8 characters
Confirm Password *
Re-enter password

Create Account Login Back

Figure 17. Alumni Registration

The screenshot displays a web interface titled "My Profile". On the left, there is a form with the following fields: "Username" (Ken), "Email" (krakenparagoso@gmail.com), "Full Name" (Kenneth Jay Paragoso A), "Program Course" (Bachelor of Science in Infor), "Year Graduated" (2026), and "Status" (approved). Below these fields are buttons for "Save Changes", "Reset", "Delete Account", and "View Survey". On the right, there is a profile section with a placeholder for a profile image, the name "Kenneth Jay Paragoso A", and the email "krakenparagoso@gmail.com". Below this is a "Profile Image" section with a "Choose File" button and "No file chosen" text, and buttons for "Upload", "Clear", and "Refresh Avatar". At the bottom right is a "Manage Password" section with fields for "Current password", "New password (min 8 chars)", and "Confirm new password", along with "Change password" and "Reset" buttons.

Figure 18. Alumni Profiling

The implementation of Figure 18 shows that the Alumni Registration and Profile Management Module has been achieved, Objective 1. The proponents confirmed that the interface is functionally suitable for alumni users, enabling individuals with minimal technical skills to easily register, log in, and update their personal and employment information.

Create digital forms to collect data.

The creation of standardized digital employment forms was able to meet the requirements of Objective 2. User assessments were completed that demonstrated the functionality and usability of the form for Alumni to input their employment information. Automated validation features are in place to increase accuracy while reducing data entry errors when using this method for recording Alumni employment compared to previous paper-based methods.

← Back

Alumni Surveys

Draft auto-saved at 11:07 PM.

A. Employment After Graduation

15. Were you Employed After Graduation? *

☒ Yes
 ☐ No

16. Was the Course degree you had in college relevant to your first Job? *

☒ Yes
 ☐ No

First Employment Details

17. Length of employment during your first employment: *

Less than six months

18. Sector of Employment in your first employment: *

Private or Non-Government Organization

19. Place of work in your first employment: *

Within Davao Oriental

20. Employment Status During your First Employment: *

Regular or Permanent

21. Gross Monthly Income (First Job): *

Below Php 13,000

23. Present Occupation: *

Technology and IT

23b. Please Specify Area of work:

web Dev

✓ Found in Bachelor of Science in Information Technology (BSIT) Professions

24. Length of Employment during your present Employment: *

Less than Six months

25. Sector of Employment in your present Employment: *

Private or Non-Government Organization

26. Place of work in your present employment: *

Within Davao Oriental

27. Employment Status during your present employment: *

Regular or Permanent

28. Gross Monthly Income (Present Job): *

Below Php 10,000

C. Job Relation

29. What competencies learned in college did you find very useful in your job? (Select 1 or More) *

- ☐ Computer Skills
- ☐ Human Relation Skills
- ☐ Interpersonal Skills
- ☐ Managerial Skills
- ☐ Entrepreneurial Skills
- ☐ Problem-Solving Skills
- ☐ Industry Skills
- ☐ Innovation
- ☐ Writing Skills

Figure 19. Collecting alumni employment information

Design role-based access controls for Alumni, Admin, and Program Heads.

As shown in Figure 20 above, the DOrSU Alumni Tracer Landing Page is the central entry point for alumni and university personnel to access all available resources of the DOrSU Alumni Tracer System. The landing page is designed with a user-friendly interface that includes tabbed navigation, an introductory paragraph describing the purpose of the system, and easily accessible “get started” buttons to help direct users through the tracing process. Because it acts as one single place where alumni and employees alike may begin their search for what they need, it allows users to quickly locate the tools and resources needed by simply clicking on it.



Figure 20. Alumni Dashboard

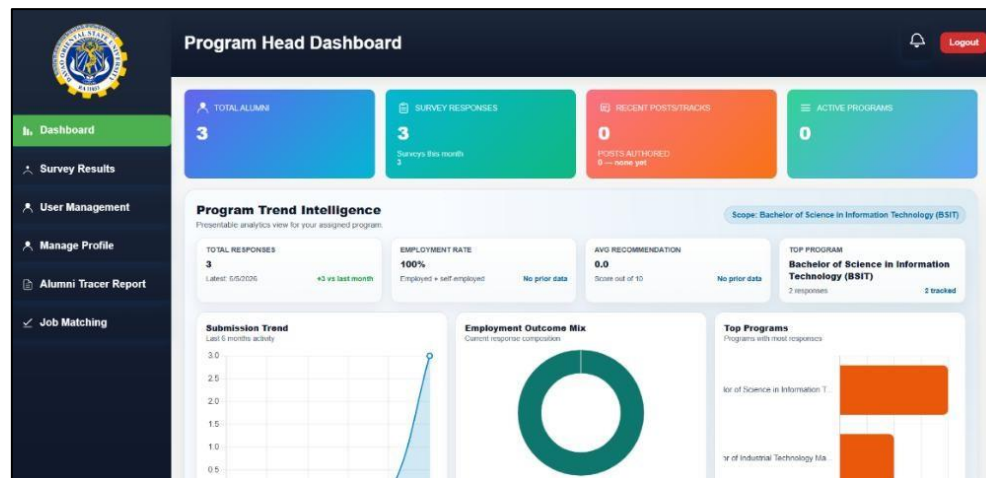


Figure 21. Program Head Dashboard

Figure 21 is shown below. The Graduate Outcome Data System's Program Head dashboard (Figure 21) is intended to be the analytic portal for academic managers to track and compare data on the graduate level. The program head can quickly view a high-level overview of the total number of alumni currently registered, the number of responses received from surveys completed by alumni during the previous month, and active programs via a clear, card-based format at the top of the screen. Beneath this are sections labeled as "program trend intelligence," where there are deeper analytical views of specific programs such as the BSIT. In addition to tracking employment rate(s), submission trends, and mix of employment outcomes related to each program, this section utilizes visual chart formats.

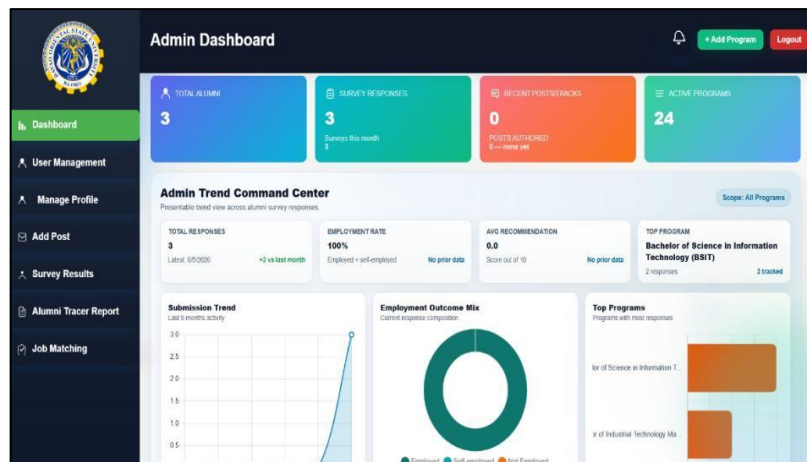


Figure 22. Admin Dashboard

Role-Based Access Control (RBAC) was integrated into the project, which met objective three. The application was very secure and provided good access control. Users could only perform actions based on their role. All alumni, admins, and program heads noted that all permissions for accessing sensitive information were correctly structured to provide safety and organization in the process.

Integrate data analytics that identify trends in employment.

Objective 4 was met through the successful deployment of the Analytics and Employment Insights Module. The system's analytics was assessed as effective and accurate, identifying trends in employment outcomes and job-degree alignment.



Figure 23. Employment Trend



Figure 24. Employed after Graduation

Develop interactive dashboards and charts.

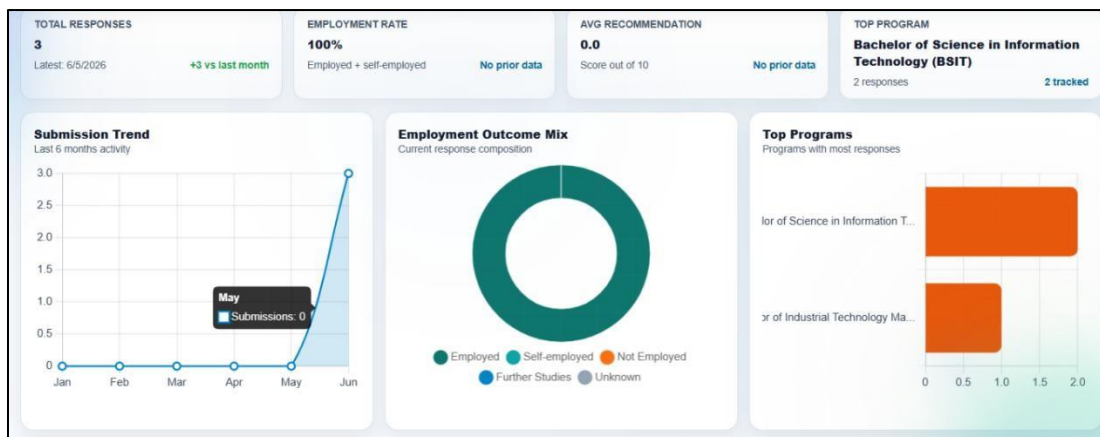


Figure 25. Visualize real-time employment patterns

Interactive dashboard implementation and real-time visualization were successful in meeting objective 5. Strong user experience feedback was received regarding usability, performance, and visual clarity on how the graphs provided a simpler way to interpret employment status, industry distribution, and relevance of job postings.

Apply rule-based logic for job-course matching algorithms.

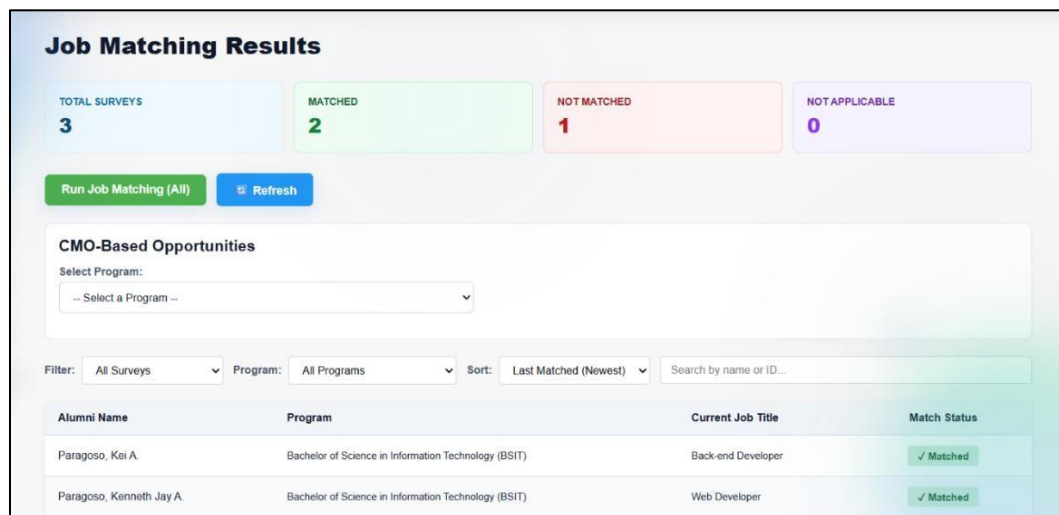


Figure 26. Matching algorithms for advanced analysis

Implementation of decision tree logic and matching algorithm met objective 6. The system was shown to be analytically correct and functionally suitable when evaluating job postings based on course relevance with an automated process. Decision-making using this new feature will help provide data-driven reviews for academic programs and enhance the system's analytic capabilities.

Provide personalized dashboards and insights

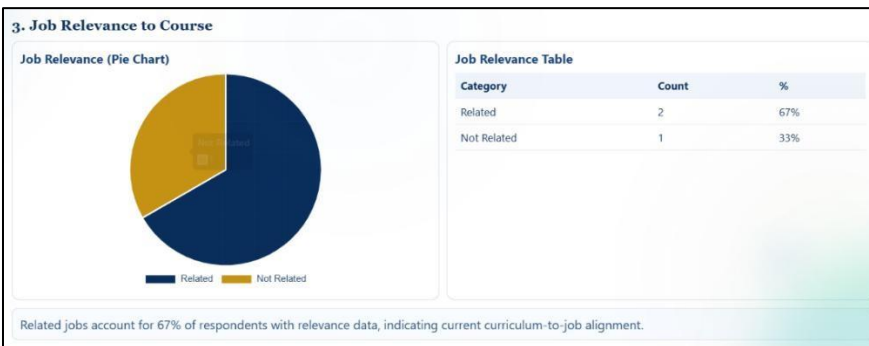
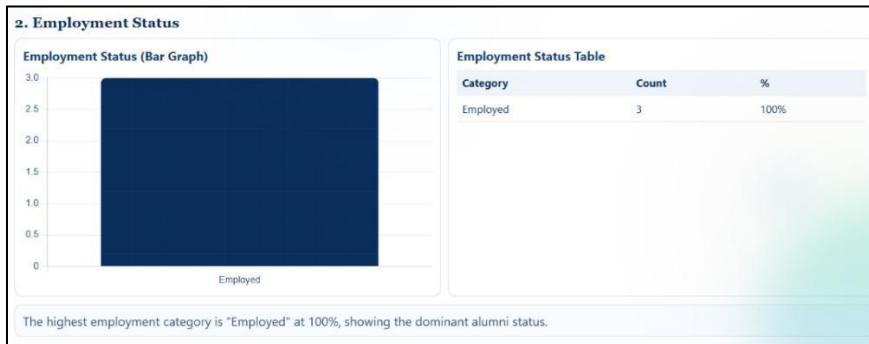




Figure 27. Personalized dashboards and insights

The creation of personalized dashboards for administrators and program heads met Objective 8. The dashboards demonstrated high usability, relevance, and functionality, giving stakeholders insights related to alumni employment outcomes. These features support evidence-based decision-making at the program and institutional levels.

Improve the accuracy, consistency, and usefulness of alumni data.

Alumni Program Head + Add User + Add Program Search ID, username, program or name Clear

Refresh Show 50 Download alumni Showing 3 of 3 alumni

Full Name	Program / Course	Year Graduate	Actions
Cheryl Iligan S	Bachelor of Early Childhood Education (BCED)	2020	View Edit Delete
Kenneth Jay Paragoso A	Bachelor of Science in Information Technology (BSIT)	2026	View Edit Delete
Kei Paragoso A	Bachelor of Science in Information Technology (BSIT)	2026	View Edit Delete

ALUMNI TRACER STUDY REPORT

Institution Name: Davao Oriental State University

Program: All Programs

Academic Year Covered: 2017 - 2026

Date of Report: June 7, 2026

Prepared By: Office of Academic Affairs

Program: All Programs Year Graduated: All Years Download Report (PDF)

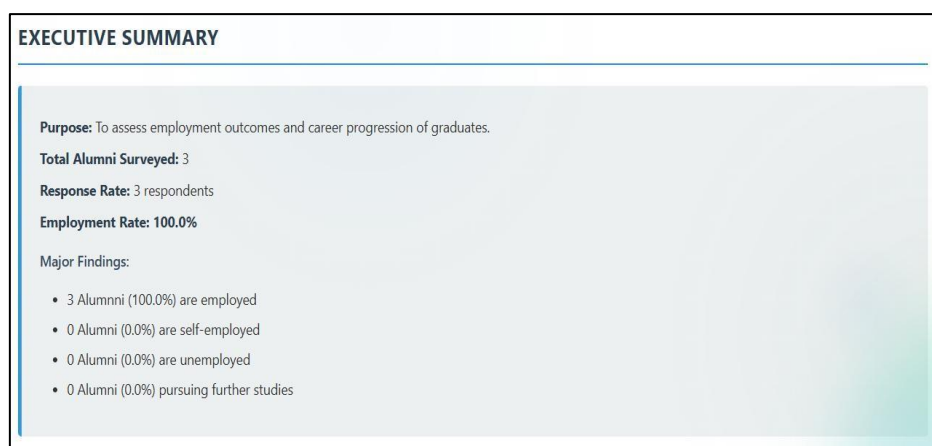


Figure 28. Alumni Tracer Report

In Figure 28, Objective 8 was fully accomplished through the system's automated reporting and data consolidation features. The reporting module met criteria for reliability, accuracy, and maintainability. Administrators confirmed that the system report generation and improved data accuracy, enabling the institution to produce consistent, high-quality alumni employment reports.

4.2 Analysis and Interpretation of User Evaluation Data

4.2.1 Evaluation Result (Admin User)

Descriptive Summary	Mean	Descriptive Level
FUNCTIONAL SUITABILITY	4.63	Strongly Agree
Functional Completeness	4.80	Strongly Agree
Functional Correctness	4.20	Agree
Functional Appropriateness	4.80	Strongly Agree
RELIABILITY	4.63	Strongly Agree
Maturity	4.80	Strongly Agree
Availability	5.00	Strongly Agree
Fault Tolerance	4.60	Strongly Agree
Recoverability	4.60	Strongly Agree
PERFORMANCE EFFICIENCY	4.60	Strongly Agree
Time Behavior	4.40	Agree
Capacity	4.40	Agree
USABILITY	4.53	Strongly Agree
Appropriateness Recognizability	5.00	Strongly Agree
Learnability	4.40	Agree
Operability	4.40	Agree
User Error Protection	4.40	Agree
SECURITY	4.80	Strongly Agree
Confidentiality	5.00	Strongly Agree
Integrity	4.40	Agree
Accountability	4.80	Strongly Agree
Authenticity	5.00	Strongly Agree
COMPATABILITY	4.56	Strongly Agree
Co-existence	4.00	Agree
Modifiability	4.40	Agree
Testability	4.40	Agree
PORTABILITY	4.60	Strongly Agree
Adaptability	4.80	Strongly Agree
Installability	4.20	Agree

Table 8. Admin Evaluation Form Result

The Functional Suitability (Alumni Tracer) of the DOrSU Alumni Tracer System received a Mean Score of 4.63, which translates into a Strongly Agree, and is representative of the administrators who agree that the system has all of the key functions

necessary to accomplish the primary goals of managing alumni data, tracking employment, and generating reports to ensure administrative accountability in these areas.

In terms of Reliability (the ability of the system to operate dependably), the system also had a Mean Score of 4.63, which was translated into a Strongly Agree; this demonstrates that it can consistently perform and operate at a stable level when operating under normal circumstances with minimal problems and providing administrators continuous access to critical alumni information.

In terms of Performance Efficiency (the speed and efficiency with which the system processes data and completes various tasks such as analytics and report generation), the system scored a Mean of 4.60, which translates into a Strongly Agree. This represents the fact that administrators view the system as quick to respond and able to handle a high volume of data processing and generate required reports quickly, regardless of how many users are accessing the system simultaneously.

Usability received a mean rating of 4.53, interpreted as Strongly Agree, demonstrating that the interface is intuitive and user-friendly. Administrators found the system easy to navigate, requiring minimal training to perform their tasks effectively.

The system excelled in Security, achieving the highest mean score of 4.80, interpreted as Strongly Agree. This highlights strong data protection mechanisms, secure authentication processes, and reliable access controls that protect sensitive alumni information.

Compatibility scored 4.56; this is interpreted as "Strongly Agree" (i.e., compatible with all university systems and workflows) and indicates that there are no conflicts when running on various operating platforms.

The system has been rated a 4.68 average score in Maintainability, which was interpreted as "Strongly Agree", indicating that the well-organized architectural design allows for relatively simple updates, debugging, and long-term maintenance/evolution of the system.

Portability rated at 4.60 and also interpreted as "Strongly Agree," demonstrates the flexibility and adaptability of the system across multiple hardware and/or software configurations. In other words, the portability of the system will allow for flexible deployment within the organizational structure of an institution.

Overall, the rating of the system was an impressive 4.68. This means that it clearly exceeds the requirements established by ISO 25010 quality standards for software and verifies that it can be used in institutions.

4.2.2 User Evaluation Form Result (Alumni User)

Table 9. Alumni Evaluation Form Result

Descriptive Summary	Mean	Descriptive Level
FUNCTIONAL SUITABILITY	4.39	Agree
Functional Completeness	4.49	Agree
Functional Correctness	4.23	Agree
Functional Appropriateness	4.60	Strongly Agree
RELIABILITY	4.46	Strongly Agree
Maturity	4.40	Agree
Availability	4.69	Strongly Agree
Fault Tolerance	4.49	Agree
Recoverability	4.63	Strongly Agree
PERFORMANCE EFFICIENCY	4.42	Agree
Time Behavior	4.57	Strongly Agree
Capacity	4.31	Agree
USABILITY	4.63	Strongly Agree
Appropriateness Recognizability	4.60	Strongly Agree
Learnability	4.29	Agree
Operability	4.43	Agree
User Error Protection	4.49	Agree
SECURITY	4.50	Strongly Agree
Confidentiality	4.83	Strongly Agree
Integrity	4.46	Agree
Accountability	4.51	Strongly Agree
Authenticity	4.71	Strongly Agree
COMPATABILITY	4.50	Strongly Agree
Co-existence	4.26	Agree
Modifiability	4.45	Agree
Testability	4.37	Agree
PORTABILITY	4.39	Agree
Adaptability	4.40	Agree
Installability	4.31	Agree

Functional Suitability was given an average rating of 4.39 (Agree). Based on this, it can be inferred that alumni are satisfied with the functionality of the system to manage their own profile and employment history, but believe there is potential for improvement by enhancing the user experience.

The Reliability had an average rating of 4.57 (Strongly Agree). This would suggest that alumni have confidence in the system's ability to consistently function while they regularly access it.

Performance Efficiency was rated an average of 4.46 (Agree), which suggests that alumni perceive the system to perform efficiently when inputting their personal data and/or retrieving required outputs.

User-friendliness (Usability) averaged 4.42 on the scale, which was interpreted as "Agree". Although alumni found the system to be user-friendly overall, it indicates there could be room for improvement in terms of improving usability by making it easier to navigate and use the system.

Security averaged 4.63 on the scale, which was interpreted as "Strongly Agree." The results show that alumni have great confidence in the system's ability to protect their personal information.

Compatibility averaged 4.5 on the scale, which was interpreted as "Strongly Agree," and indicates that the system operates effectively across most types of computer hardware and software platforms used by alumni.

Maintainability also averaged 4.5 on the scale, which was interpreted as "Strongly Agree", and indicated that alumni recognized the system had been designed with longevity and updatable features.

Finally, Portability averaged 4.39 on the scale, which was interpreted as "Agree," and confirmed that the system has adequate flexibility to function in other than the original operating environment.

Overall, the system received an impressive rating of 4.50. This shows that it fully meets ISO 25010 quality standards and confirms that it is ready for use in institutions.

CHAPTER V

SUMMARY, CONCLUSION, AND RECOMMENDATIONS

5.1 Summary

The DOrSU Alumni Tracer System is an innovation in the way of monitoring graduates for Davao Oriental State University. The DOrSU Alumni Tracer System will replace traditional methods of alumni tracking by using digital information collection, auto-analyzing that data, and providing interactive reporting features. The DOrSU Alumni Tracer System will allow for the effective and timely monitoring of a student's career advancement through alumni profile creation and management, employment data collection, and smart job/course pairing utilizing rule-based algorithms.

The online alumni platform is designed to be a better alternative to handling all of the tasks manually. This was achieved by providing tools that include automated report creation, real-time dashboards, and role-based access to information. Because it is cloud-hosted, this system provides a secure way to store, manage, and provide access to your institution's data from any device. This allows the institution to improve communication with its alumni (and potentially other constituents), increase data integrity, and use data analysis to create new programs, enhance curriculum, and develop plans for future development.

Implementation results show the system's effectiveness, with administrators rating it as Strongly Agree across all ISO 25010 quality standards. The platform modernizes graduate tracking procedures, supports better decision-making for enhancing academic programs, and strengthens the bond between the university and its alumni.

5.2 Conclusion

In addition to providing a model for tracking graduates from alumni programs, the DOrSU Alumni Tracer System represents an example of how technology is changing the way institutions collect data about their students' post-graduation experiences. The tracer system's digital format provides many benefits over its predecessor (a paper-based system), such as increased data integrity, more efficient operations, and greater user engagement. Additionally, features like online participant profiles, automated analysis capabilities, and interactive reporting tools provide institutional researchers with the means to monitor participants' career development while also assessing the relevance of their respective academic programs.

This project demonstrates how technology can bridge gaps between schools and their graduates, as well as assist with data-driven decision-making. The system will not only be able to make administrative functions easier but also strengthen the connection that is so important between an institution of higher learning and its alumni community. The success of the DOrSU Alumni Tracer System has established a new level of tracking graduates and shows how innovation using technology can increase both the efficiency and effectiveness of institutions of higher education, supporting their graduates.

5.3 Recommendation

The following recommendations are given for future researchers based on the study's findings and conclusions:

- Analyze AI/ML as possible sources for forecasting job market trends and updated professional guidance (career advice) for alumni.

- Focus your research on developing an application that is accessible via mobile device(s), using cross-platform design, and that operates when there is no Internet connection. This could greatly enhance user accessibility in those areas where there are limited or no Internet services available.
- Develop and integrate enhanced security features, including blockchain-based systems, to ensure the accuracy/integrity of alumni data and multi-factor authentication systems to protect confidential/sensitive alumni data.

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APPENDICES

APPENDIX A.

List of request and approval to conduct the study

Republic of the Philippines

DAVAO ORIENTAL STATE UNIVERSITY

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Date: February 17, 2025

MR. JONATHAN S. CABRERA

Faculty – BSIT Program
Institute of Computing and Engineering
This State University

Dear Mr. Cabrera,

Greetings of Peace!

We, the BSIT students of this University namely, **Mike Misoles, Kristine Mae Bonotan, Kenneth Jay Paragoso** would like to submit our intent to pursue with the Capstone Project entitled “**DORSU Alumni Tracer: A Data-Driven Approach to Monitoring Alumni Career Progress**”. In this regard, may we request through this communication letter that you will be our study adviser for this project as we believed that your knowledge and expertise on this field/area of study will be valuable and will greatly help us enrich this project.

Thank you for your consideration, and we are hoping for your positive response with regards to this request. Attached herewith is the concept paper submitted and duly approved by our Capstone Project coordinator for your perusal.

Respectfully Yours,

MIKE MISOLE

(Signature over printed name)

KRISTINE MAE BONOTAN

(Signature over printed name)

KENNETH JAY PARAGOSO

(Signature over printed name)

The Proponents

Noted by:

MS. CINDY LASCO

Capstone Project Coordinator

RESPONSE

(Please check one for your response.)



Yes, I agree to be your study adviser.



I do not agree. Kindly look for another study adviser due to the reason indicated below

Reason:

Interesting

(Signature over printed name)

Davao Oriental State University
Guang-guang, Dahican, City of Mati,
Davao Oriental, 8200
Republic of the Philippines

website: www.dorsu.edu.ph
phone: +63 (087) 3883 195
e-mail: op@dorsu.edu.ph
Facebook: @dorsuofficial



APPENDIX B.

Routing Slip

APPENDIX C.

Grammarly Report

Bonotan et. al CAPSTONE2.docx

 Capitol University

Document Details

Submission ID

trn:oid::30916:143565751

Submission Date

Jun 19, 2026, 10:39 AM GMT+8

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File Name

Bonotan et. al CAPSTONE2.docx

File Size

7.2 MB

77 Pages

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Page 1 of 84 - Cover Page

Submission ID trn:oid::30916:143565751



Page 2 of 84 - Integrity Overview

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APPENDIX D.

Grammarian's Certification

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CERTIFICATION

This is to certify that the capstone project titled **"DORSU ALUMNI TRACER: A DATA-DRIVEN APPROACH TO MONITORING ALUMNI CAREER PROGRESS"** authored by **Kenneth Jay Paragoso, Mike Misoles, and Kristine Mae Bonotan**, has been thoroughly reviewed for grammar, syntax, punctuation, and overall language use.

The manuscript has been carefully examined to ensure clarity, coherence, and adherence to academic writing conventions. All necessary grammatical corrections have been recommended and/or applied to uphold the quality and readability of the work.

Given this 19th of June 2026 at the Davao Oriental State University- Main Campus.

Certified By:


Loremie L. Masunag, LPT
Grammarian

APPENDIX E.

Expert Evaluation Form

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November 24, 2025

RODRIGO AMPER SALIMACO JR., PhD.
BSM Faculty
Davao Oriental State University

Dear Mr. Salimaco,

Good day! We are conducting a research study titled **"DORSU ALUMNI TRACER: A DATA-DRIVEN APPROACH TO MONITORING ALUMNI CAREER PROGRESS"** at Davao Oriental State University. This study aims to create a **website** that will track the careers of our graduates. The goal is to gather data that will help improve university programs and student success.

As part of our study, we have prepared a survey instrument that evaluates the system's quality across eight domains based on ISO 25010 standards: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. We respectfully request your expertise in validating this instrument by assessing the degree of relevance and clarity of each item.

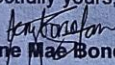
Please use the following rating scale as your guide:

Degree of Relevance:

- 1 = The item is not relevant.
- 2 = The item is somewhat relevant.
- 3 = The item is quite relevant.
- 4 = The item is highly relevant.

Thank you very much for your time and expertise. More power and God bless.

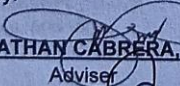
Respectfully yours,

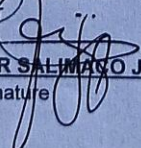

Kristine Mae Bonotan
Researcher


Mike Misoles
Researcher


Kenneth Jay Paragoso
Researcher

Noted by:


JONATHAN CABRERA, PhD
Adviser


RODRIGO AMPER SALIMACO JR., PhD.
Signature

Assistant Professor II
Expert's Title/Position

Date Signed: _____

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Date Signed: _____

Instruction					
Please indicate to what extent you agree on the given statement.					
ISO/IEC 25010 standards for software quality evaluation used					Comment
1 – Strongly Disagree; 2 – Disagree; 3 – Neutral; 4 – Agree; 5 – Strongly Agree					
Domain 1:	Your Assessment (Kindly check the column which is appropriate for each item)				
Functional Suitability	Relevance				
	(1) No relevance at all	(2) Item is somewhat relevant	(3) Item is quite relevant	(4) Item is highly relevant	
1. Functional Completeness The DORSU Alumni Tracer provides the features needed to track alumni data such as employment status, job titles, and further studies.				/	
2. Functional Completeness The alumni information and reports in the system can be displayed, filtered, and sorted as needed.				/	
3. Functional Correctness The website's menus, buttons, and forms respond correctly when used.				/	
4. Functional Correctness The generated alumni reports contain accurate and complete information.				/	

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5. Functional Appropriateness The information presented for each alumni profile is complete and relevant.			/		
6. Functional Appropriateness The charts, graphs, and statistics make the alumni career data easy to understand and analyze.			/		
Instruction					Comment
Please indicate to what extent you agree on the given statement.					
ISO/IEC 25010 standards for software quality evaluation used					Comment
1 – Strongly Disagree; 2 – Disagree; 3 – Neutral; 4 – Agree; 5 – Strongly Agree					
Domain 2: Reliability	Your Assessment (Kindly check the column which is appropriate for each item)				
	Relevance				
	(1) No relevance at all	(2) Item is somewhat relevant	(3) Item is quite relevant	(4) Item is highly relevant	
1. Maturity DORSU Alumni Tracer website works without errors or crashes during use.				/	
2. Availability Alumni tracer website can be accessed anytime when needed.				/	
3. Availability The website loads successfully and is ready to use without long delays.				/	

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4. Fault Tolerance When an error occurs (e.g., a failed login or search), the system allows users to try again instead of crashing.						
5. Recoverability After refreshing the page or logging back in, user data and session information are restored properly.						
6. Recoverability The website continues working normally after a user's session has expired and they log in again.						
Instruction						Comment
Please indicate to what extent you agree on the given statement.						
ISO/IEC 25010 standards for software quality evaluation used						Comment
1 – Strongly Disagree; 2 – Disagree; 3 – Neutral; 4 – Agree; 5 – Strongly Agree						
Domain 3: Performance Efficiency	Your Assessment (Kindly check the column which is appropriate for each item)					
	Relevance					
	(1) No relevance at all	(2) Item is somewhat relevant	(3) Item is quite relevant	(4) Item is highly relevant		
1. Time Behavior DORSU Alumni Tracer website loads and becomes ready to use without noticeable delays.						
2. Time Behavior The charts, graphs, and statistical reports are generated and displayed smoothly after a search or data request.						
3. Time Behavior The website responds quickly when navigating between different sections, such as the dashboard, alumni list,						

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and reports.					
4. Resource Utilization The system runs smoothly in a web browser without lagging or freezing.					
5. Resource Utilization The website does not consume excessive memory or processing power from the user's device while in use.					
6. Capacity The website still performs well and remains stable even during periods of high traffic from many simultaneous users.					
Instruction					Comment
Please indicate to what extent you agree on the given statement.					
ISO/IEC 25010 standards for software quality evaluation used					Comment
1 – Strongly Disagree; 2 – Disagree; 3 – Neutral; 4 – Agree; 5 – Strongly Agree					
Domain 4:	Your Assessment				
Usability	(Kindly check the column which is appropriate for each item)				
	Relevance				
	(1) No relevance at all	(2) Item is somewhat relevant	(3) Item is quite relevant	(4) Item is highly relevant	
1. Appropriateness Recognizability It is easy for users to understand the website's purpose and what tasks they can accomplish with it.					
2. Learnability It is easy to learn how to navigate the website and use its features, even without prior instructions.					

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3. Operability The website's buttons, menus, and navigation controls are intuitive and simple to use.					
4. User Error Protection The system shows clear warning messages and confirmations to help prevent user errors, such as submitting incomplete forms.					
5. User Interface Aesthetics The website's design, color scheme, and overall layout are visually appealing and professional.					
6. Accessibility The website can be used effectively by different alumni, including those with varying levels of technical skill.					
Instruction					Comment
Please indicate to what extent you agree on the given statement.					
ISO/IEC 25010 standards for software quality evaluation used					Comment
1 – Strongly Disagree; 2 – Disagree; 3 – Neutral; 4 – Agree; 5 – Strongly Agree					
Domain 5: Security	Your Assessment (Kindly check the column which is appropriate for each item)				
	Relevance				
	(1) No relevance at all	(2) Item is somewhat relevant	(3) Item is quite relevant	(4) Item is highly relevant	
1. Confidentiality User login information and personal profile data are kept private and secure.					

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2. Integrity The system prevents unauthorized access to ensure that alumni data cannot be viewed or modified by unapproved individuals.					
3. Accountability The system ensures that all new user accounts are verified and approved by an administrator before access is granted.					
4. Authenticity Only approved and authenticated program head users can access the alumni tracer system administration.					

Instruction	Comment
-------------	---------

Please indicate to what extent you agree on the given statement.

ISO/IEC 25010 standards for software quality evaluation used	Comment
--	---------

1 – Strongly Disagree; 2 – Disagree; 3 – Neutral; 4 – Agree; 5 – Strongly Agree

Domain 6: Compatibility	Your Assessment (Kindly check the column which is appropriate for each item)				
	Relevance				
	(1) No relevance at all	(2) Item is somewhat relevant	(3) Item is quite relevant	(4) Item is highly relevant	
1. Co-existence The website functions smoothly even when other browser tabs or applications are open.					
2. Co-existence The website operates efficiently without causing the web browser or the device to slow down or freeze.					

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3. Co-existence The website runs properly within the browser without conflicts from browser extensions or other software.						
4. Interoperability The website works correctly on different web browsers.						
5. Interoperability The website displays and functions correctly on various devices.						
Instruction						Comment
Please indicate to what extent you agree on the given statement.						
ISO/IEC 25010 standards for software quality evaluation used						Comment
1 – Strongly Disagree; 2 – Disagree; 3 – Neutral; 4 – Agree; 5 – Strongly Agree						
Domain 7:	Your Assessment (Kindly check the column which is appropriate for each item)					
Maintainability	Relevance					
	(1) No relevance at all	(2) Item is somewhat relevant	(3) Item is quite relevant	(4) Item is highly relevant		
1. Modularity The website remains stable and performs consistently, even during extended use.						
2. Reusability The system's framework and features are adaptable and could be repurposed for other university tracer studies or data-monitoring tasks.						
3. Analyzability If an error occurs, the system provides clear and understandable error messages that help identify what went wrong.						

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4. Modifiability The website's design makes it seem straightforward to add new features or update existing ones in the future.					
5. Testability The website feels well-built, functioning as intended with very few bugs or errors.					
Instruction	Comment				
Please indicate to what extent you agree on the given statement.					
ISO/IEC 25010 standards for software quality evaluation used					Comment
1 – Strongly Disagree; 2 – Disagree; 3 – Neutral; 4 – Agree; 5 – Strongly Agree					
Domain 8: Portability	Your Assessment (Kindly check the column which is appropriate for each item)				
Relevance (1) No relevance at all (2) Item is somewhat relevant (3) Item is quite relevant (4) Item is highly relevant					
1. Adaptability The website works properly on different versions of web browsers and operating systems without any issues.					
2. Installability It was easy for me to access and set up an account on the website.					
3. Replaceability This system can easily replace other methods of alumni tracking without causing problems or loss of historical data.					

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November 24, 2025

VINZ LLOYD A. ALFEREZ
BSIT Faculty
Davao Oriental State University

Dear Mr. Alferez,

Good day! We are conducting a research study titled "**DORSU ALUMNI TRACER: A DATA-DRIVEN APPROACH TO MONITORING ALUMNI CAREER PROGRESS**" at Davao Oriental State University. This study aims to create a **website** that will track the careers of our graduates. The goal is to gather data that will help improve university programs and student success.

As part of our study, we have prepared a survey instrument that evaluates the system's quality across eight domains based on ISO 25010 standards: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. We respectfully request your expertise in validating this instrument by assessing the degree of relevance and clarity of each item.

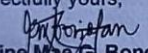
Please use the following rating scale as your guide:

Degree of Relevance:

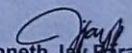
- 1 = The item is not relevant.
- 2 = The item is somewhat relevant.
- 3 = The item is quite relevant.
- 4 = The item is highly relevant.

Thank you very much for your time and expertise. More power and God bless.

Respectfully yours,


Kristine Mae G. Bonotan
Researcher

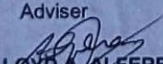

Mike Misoles
Researcher


Kenneth J. Paragoso
Researcher

Noted by:


JONATHAN CABRERA, PhD

Adviser


VINZ LLOYD A. ALFEREZ
Signature

Instructor I
Expert's Title/Position

Date Signed: _____

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Date Signed: _____

Instruction					
Please indicate to what extent you agree on the given statement.					
ISO/IEC 25010 standards for software quality evaluation used					Comment
1 – Strongly Disagree; 2 – Disagree; 3 – Neutral; 4 – Agree; 5 – Strongly Agree					
Domain 1: Functional Suitability	Your Assessment (Kindly check the column which is appropriate for each item)				
	Relevance				
	(1) No relevance at all	(2) Item is somewhat relevant	(3) Item is quite relevant	(4) Item is highly relevant	
1. Functional Completeness The DORSU Alumni Tracer provides the features needed to track alumni data such as employment status, job titles, and further studies.				/	
2. Functional Completeness The alumni information and reports in the system can be displayed, filtered, and sorted as needed.				/	
3. Functional Correctness The website's menus, buttons, and forms respond correctly when used.			/		
4. Functional Correctness The generated alumni reports contain accurate and complete information.					

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5. Functional Appropriateness The information presented for each alumni profile is complete and relevant.					
6. Functional Appropriateness The charts, graphs, and statistics make the alumni career data easy to understand and analyze.					
Instruction					Comment
Please indicate to what extent you agree on the given statement.					
ISO/IEC 25010 standards for software quality evaluation used					Comment
1 – Strongly Disagree; 2 – Disagree; 3 – Neutral; 4 – Agree; 5 – Strongly Agree					
Domain 2: Reliability	Your Assessment (Kindly check the column which is appropriate for each item)				
	Relevance				
	(1) No relevance at all	(2) Item is somewhat relevant	(3) Item is quite relevant	(4) Item is highly relevant	
1. Maturity DORSU Alumni Tracer website works without errors or crashes during use.					Change to unexpected errors
2. Availability Alumni tracer website can be accessed anytime when needed.					
3. Availability The website loads successfully and is ready to use without long delays.					

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4. Fault Tolerance When an error occurs (e.g., a failed login or search), the system allows users to try again instead of crashing.				/	
5. Recoverability After refreshing the page or logging back in, user data and session information are restored properly.				/	
6. Recoverability The website continues working normally after a user's session has expired and they log in again.				/	
Instruction					Comment
Please indicate to what extent you agree on the given statement.					
ISO/IEC 25010 standards for software quality evaluation used					Comment
1 – Strongly Disagree; 2 – Disagree; 3 – Neutral; 4 – Agree; 5 – Strongly Agree					
Domain 3:	Your Assessment				
Performance Efficiency	(Kindly check the column which is appropriate for each item)				
	Relevance				
	(1) No relevance at all	(2) Item is somewhat relevant	(3) Item is quite relevant	(4) Item is highly relevant	
1. Time Behavior DORSU Alumni Tracer website loads and becomes ready to use without noticeable delays.			/		
2. Time Behavior The charts, graphs, and statistical reports are generated and displayed smoothly after a search or data request.		/			
3. Time Behavior The website responds quickly when navigating between different sections, such as the dashboard, alumni list,			/		

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and reports.

4. Resource Utilization
The system runs smoothly in a web browser without lagging or freezing.

5. Resource Utilization
The website does not consume excessive memory or processing power from the user's device while in use.

6. Capacity
The website still performs well and remains stable even during periods of high traffic from many simultaneous users.

Instruction Comment

Please indicate to what extent you agree on the given statement.

ISO/IEC 25010 standards for software quality evaluation used Comment

1 – Strongly Disagree; 2 – Disagree; 3 – Neutral; 4 – Agree; 5 – Strongly Agree

Domain 4:

Usability

Your Assessment

(Kindly check the column which is appropriate for each item)

Relevance

(1) No relevance at all	(2) Item is somewhat relevant	(3) Item is quite relevant	(4) Item is highly relevant

**1. Appropriateness
Recognizability**
It is easy for users to understand the website's purpose and what tasks they can accomplish with it.

2. Learnability
It is easy to learn how to navigate the website and use its features, even without prior instructions.

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3. Operability The website's buttons, menus, and navigation controls are intuitive and simple to use.						
4. User Error Protection The system shows clear warning messages and confirmations to help prevent user errors, such as submitting incomplete forms.						
5. User Interface Aesthetics The website's design, color scheme, and overall layout are visually appealing and professional.						
6. Accessibility The website can be used effectively by different alumni, including those with varying levels of technical skill.						
Instruction						Comment
Please indicate to what extent you agree on the given statement.						
ISO/IEC 25010 standards for software quality evaluation used						Comment
1 – Strongly Disagree; 2 – Disagree; 3 – Neutral; 4 – Agree; 5 – Strongly Agree						
Domain 5: Security	Your Assessment (Kindly check the column which is appropriate for each item)					
	Relevance					
	(1) No relevance at all	(2) Item is somewhat relevant	(3) Item is quite relevant	(4) Item is highly relevant		
1. Confidentiality User login information and personal profile data are kept private and secure.						

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2. Integrity The system prevents unauthorized access to ensure that alumni data cannot be viewed or modified by unapproved individuals.						
3. Accountability The system ensures that all new user accounts are verified and approved by an administrator before access is granted.						
4. Authenticity Only approved and authenticated program head users can access the alumni tracer system administration.						
Instruction						Comment
Please indicate to what extent you agree on the given statement.						
ISO/IEC 25010 standards for software quality evaluation used						Comment
1 – Strongly Disagree; 2 – Disagree; 3 – Neutral; 4 – Agree; 5 – Strongly Agree						
Domain 6:	Your Assessment					
Compatibility	(Kindly check the column which is appropriate for each item)					
	Relevance					
	(1) No relevance at all	(2) Item is somewhat relevant	(3) Item is quite relevant	(4) Item is highly relevant		
1. Co-existence The website functions smoothly even when other browser tabs or applications are open.						
2. Co-existence The website operates efficiently without causing the web browser or the device to slow down or freeze.						

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3. Co-existence The website runs properly within the browser without conflicts from browser extensions or other software.						
4. Interoperability The website works correctly on different web browsers.						
5. Interoperability The website displays and functions correctly on various devices.						
Instruction						Comment
Please indicate to what extent you agree on the given statement.						
ISO/IEC 25010 standards for software quality evaluation used						Comment
1 – Strongly Disagree; 2 – Disagree; 3 – Neutral; 4 – Agree; 5 – Strongly Agree						
Domain 7:	Your Assessment					
Maintainability	(Kindly check the column which is appropriate for each item)					
	Relevance					
	(1) No relevance at all	(2) Item is somewhat relevant	(3) Item is quite relevant	(4) Item is highly relevant		
1. Modularity The website remains stable and performs consistently, even during extended use.						
2. Reusability The system's framework and features are adaptable and could be repurposed for other university tracer studies or data-monitoring tasks.						
3. Analyzability If an error occurs, the system provides clear and understandable error messages that help identify what went wrong.						

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4. Modifiability The website's design makes it seem straightforward to add new features or update existing ones in the future.					
5. Testability The website feels well-built, functioning as intended with very few bugs or errors.					
Instruction					Comment
Please indicate to what extent you agree on the given statement.					
ISO/IEC 25010 standards for software quality evaluation used					Comment
1 – Strongly Disagree; 2 – Disagree; 3 – Neutral; 4 – Agree; 5 – Strongly Agree					
Domain 8: Portability	Your Assessment (Kindly check the column which is appropriate for each item)				
	Relevance				
	(1) No relevance at all	(2) Item is somewhat relevant	(3) Item is quite relevant	(4) Item is highly relevant	
1. Adaptability The website works properly on different versions of web browsers and operating systems without any issues.					
2. Installability It was easy for me to access and set up an account on the website.					
3. Replaceability This system can easily replace other methods of alumni tracking without causing problems or loss of historical data.					

APPENDIX F.

Survey User Evaluation Questionnaires

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Name (optional): _____

Age: _____

Affiliated Institution: _____

Position/Designation: _____

PART 2: SOFTWARE QUALITY EVALUATION

A. **Directions:** Please evaluate the software quality of "DORSU ALUMNI TRACER: A DATA-DRIVEN APPROACH TO MONITORING ALUMNI CAREER PROGRESS" based on the eight characteristics of ISO 25010 Software Product Quality Standards for end users. Check the appropriate box using the five-point scale with its corresponding verbal descriptions (5 as the highest and 1 as the lowest).

Functional Suitability – represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions.

Questions	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
The DORSU Alumni Tracer provides the features needed to track alumni data, including employment status, job titles, and further studies.					
The alumni information and reports in the system can be displayed, filtered, and sorted as needed.					
The website's menus, buttons, and forms respond correctly when used.					
The generated alumni reports contain accurate and complete information.					
The information presented for each alumni profile is complete and relevant.					
The charts, graphs, and statistics make the alumni career data easy to understand and analyze.					

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Reliability – represents the degree to which a product or system performs its required functions without failure under stated conditions for a specified period of time.

Questions	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
The DOrSU Alumni Tracer website works without unexpected errors during use.					
The alumni tracer website can be accessed anytime it's needed.					
The website loads successfully and is ready to use without long delays.					
When an error occurs (e.g., a failed login or search), the system allows users to try again instead of crashing.					
After refreshing the page or logging back in, user data and session information are restored properly.					
The website continues working normally after a user's session has expired and they log in again.					

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Performance Efficiency – represents the degree to which a system performs its functions with optimal resource usage and speed under specified conditions.

Questions	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
DORSU Alumni Tracer website loads and becomes ready to use without noticeable delays.					
The charts, graphs, and statistical reports are generated and displayed smoothly after a search or data request.					
The website responds quickly when navigating between different sections, such as the dashboard, alumni list, and reports.					
The system runs smoothly in a web browser without lagging or freezing.					
The website does not consume excessive memory or processing power from the user's device while in use.					
The website still performs well and remains stable even during periods of high traffic from many simultaneous users.					

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Usability – represents the degree to which a system can be used efficiently, easily, and satisfactorily by users to achieve their goals under specified conditions.

Questions	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
It is easy for users to understand the website's purpose and what tasks they can accomplish with it.					
It is easy to learn how to navigate the website and use its features, even without prior instructions.					
The website's buttons, menus, and navigation controls are intuitive and simple to use.					
The system shows clear warning messages and confirmations to help prevent user errors, such as submitting incomplete forms.					
The website's design, color scheme, and overall layout are visually appealing and professional.					
The website can be used effectively by different alumni, including those with varying levels of technical skill.					

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Security – represents the degree to which a system protects data and operations against unauthorized access, misuse, or harm under specified conditions.

Questions	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
User login information and personal profile data are kept private and secure.					
The system prevents unauthorized access to ensure that alumni data cannot be viewed or modified by unapproved individuals.					
The system ensures that all new user accounts are verified and approved by an administrator before access is granted.					
Only approved and authenticated program head users can access the alumni tracer system administration.					

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Compatibility – represents the degree to which Alumni Tracer can operate alongside other applications and be used across different platforms, browsers, and devices under specified conditions.

Questions	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
The website functions smoothly even when other browser tabs or applications are open.					
The website operates efficiently without causing the web browser or the device to slow down or freeze.					
The website runs properly within the browser without conflicts from browser extensions or other software.					
The website works correctly on different web browsers.					
The website displays and functions correctly on various devices.					

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Maintainability – represents the degree to which Alumni Tracer can be analyzed, modified, tested, and maintained by technical personnel under specified conditions.

Questions	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
The website remains stable and performs consistently, even during extended use.					
The system's framework and features are adaptable and could be repurposed for other university tracer studies or data-monitoring tasks.					
If an error occurs, the system provides clear and understandable error messages that help identify what went wrong.					
The website's design makes it seem straightforward to add new features or update existing ones in the future.					
The website feels well-built, functioning as intended with very few bugs or errors.					

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Portability – represents the degree to which a system can be easily transferred, installed, and used across different environments or platforms under specified conditions.

Questions	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
The website works properly on different versions of web browsers and operating systems without any issues.					
It was easy for me to access and set up an account on the website.					
This system can easily replace other methods of alumni tracking without causing problems or loss of historical data.					

APPENDIX G.

Sample filled-out User Evaluation Questionnaires

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Questionnaire for End User

PART 1: RESPONDENT'S PROFILE

Name: JACIN DARE P. DORINO

Age: 22

Affiliated Institution: FALU

Position/Designation: STUDENT

PART 2: SOFTWARE QUALITY EVALUATION

A. Directions: Please evaluate the software quality of **"DORSU ALUMNI TRACER: A DATA-DRIVEN APPROACH TO MONITORING ALUMNI CAREER PROGRESS"** based on the eight characteristics of ISO 25010 Software Product Quality Standards for end users. Check the appropriate box using the five-point scale with its corresponding verbal descriptions (5 as the highest and 1 as the lowest).

Functional Suitability – represent the degree to which a product or system provides functions that meet stated and implied needs when used under specified condition.

Questions	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
The DORSU Alumni Tracer provides the features needed to track alumni data such as employment status, job titles, and further studies.					/
The alumni information and reports in the system can be displayed, filtered, and sorted as needed.				/	
The website's menus, buttons, and forms respond correctly when used.					/
The generated alumni reports contain accurate and complete information.				/	
The information presented for each alumni profile is complete and relevant.					/
The charts, graphs, and statistics make the alumni career data easy to understand and analyze.				/	

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Reliability – represent the degree to which a product or system performs its required functions without failure under stated conditions for a specified period of time.

Questions	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
The DOrSU Alumni Tracer website works without errors or crashes during use.				/	
Alumni tracer website can be accessed anytime when needed.					/
The website loads successfully and is ready to use without long delays.				/	
When an error occurs (e.g., a failed login or search), the system allows users to try again instead of crashing.				/	
After refreshing the page or logging back in, user data and session information are restored properly.				/	
The website continues working normally after a user's session has expired and they log in again.					/

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Performance Efficiency – represents the degree to which a system performs its functions with optimal resource usage and speed under specified conditions.

Questions	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
DORSU Alumni Tracer website loads and becomes ready to use without noticeable delays.				/	
The charts, graphs, and statistical reports are generated and displayed smoothly after a search or data request.					/
The website responds quickly when navigating between different sections, such as the dashboard, alumni list, and reports.					/
The system runs smoothly in a web browser without lagging or freezing.				/	
The website does not consume excessive memory or processing power from the user's device while in use.					/
The website still performs well and remains stable even during periods of high traffic from many simultaneous users.					/

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Usability – represents the degree to which a system can be used efficiently, easily, and satisfactorily by users to achieve their goals under specified conditions.

Questions	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
It is easy for users to understand the website's purpose and what tasks they can accomplish with it.					/
It is easy to learn how to navigate the website and use its features, even without prior instructions.					/
The website's buttons, menus, and navigation controls are intuitive and simple to use.					/
The system shows clear warning messages and confirmations to help prevent user errors, such as submitting incomplete forms.					/
The website's design, color scheme, and overall layout are visually appealing and professional.				/	
The website can be used effectively by different alumni, including those with varying levels of technical skill.					/

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Questions	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
User login information and personal profile data are kept private and secure.					/
The system prevents unauthorized access to ensure that alumni data cannot be viewed or modified by unapproved individuals.					/
The system ensures that all new user accounts are verified and approved by an administrator before access is granted.					/
Only approved and authenticated program head users can access the alumni tracer system administration.					/

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Compatibility – represents the degree to which CrimeCast can operate alongside other applications and be used across different platforms, browsers, and devices under specified conditions.

Questions	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
The website functions smoothly even when other browser tabs or applications are open.					/
The website operates efficiently without causing the web browser or the device to slow down or freeze.					/
The website runs properly within the browser without conflicts from browser extensions or other software.				/	
The website works correctly on different web browsers.					/
The website displays and functions correctly on various devices.				/	

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Questions	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
The website remains stable and performs consistently, even during extended use.				/	
The system's framework and features are adaptable and could be repurposed for other university tracer studies or data-monitoring tasks. entially be adapted for use in other cities.					/
If an error occurs, the system provides clear and understandable error messages that help identify what went wrong.					/
The website's design makes it seem straightforward to add new features or update existing ones in the future.					/
The website feels well-built, functioning as intended with very few bugs or errors.				/	

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Questions	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
The website works properly on different versions of web browsers and operating systems without any issues.				/	
It was easy for me to access and set up an account on the website.				/	
This system can easily replace other methods of alumni tracking without causing problems or loss of historical data.				/	

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Questionnaire for End User

PART 1: RESPONDENT'S PROFILE

Name: Kevin R. Ortiz
 Age: 19
 Affiliated Institution: FALC
 Position/Designation: Student

PART 2: SOFTWARE QUALITY EVALUATION

A. **Directions:** Please evaluate the software quality of "**DORSU ALUMNI TRACER: A DATA-DRIVEN APPROACH TO MONITORING ALUMNI CAREER PROGRESS**" based on the eight characteristics of ISO 25010 Software Product Quality Standards for end users. Check the appropriate box using the five-point scale with its corresponding verbal descriptions (5 as the highest and 1 as the lowest).

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Questions	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
The DORSU Alumni Tracer provides the features needed to track alumni data such as employment status, job titles, and further studies.				✓	
The alumni information and reports in the system can be displayed, filtered, and sorted as needed.				✓	
The website's menus, buttons, and forms respond correctly when used.			✓		
The generated alumni reports contain accurate and complete information.				✓	
The information presented for each alumni profile is complete and relevant.					✓
The charts, graphs, and statistics make the alumni career data easy to understand and analyze.				✓	

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Alumni tracer website can be accessed anytime when needed.				/	
The website loads successfully and is ready to use without long delays.					/
When an error occurs (e.g., a failed login or search), the system allows users to try again instead of crashing.				/	
After refreshing the page or logging back in, user data and session information are restored properly.				/	
The website continues working normally after a user's session has expired and they log in again.			/		

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The website's buttons, menus, and navigation controls are intuitive and simple to use.					/
The system shows clear warning messages and confirmations to help prevent user errors, such as submitting incomplete forms.				/	
The website's design, color scheme, and overall layout are visually appealing and professional.				/	
The website can be used effectively by different alumni, including those with varying levels of technical skill.					/

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The system prevents unauthorized access to ensure that alumni data cannot be viewed or modified by unapproved individuals.					/
The system ensures that all new user accounts are verified and approved by an administrator before access is granted.					/
Only approved and authenticated program head users can access the alumni tracer system administration.				/	

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The website operates efficiently without causing the web browser or the device to slow down or freeze.				/	
The website runs properly within the browser without conflicts from browser extensions or other software.					/
The website works correctly on different web browsers.				/	
The website displays and functions correctly on various devices.				/	

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The website works properly on different versions of web browsers and operating systems without any issues.				/	
It was easy for me to access and set up an account on the website.					/
This system can easily replace other methods of alumni tracking without causing problems or loss of historical data.				/	

APPENDIX H.

Statistician's Certificate

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Document Code No.			
FM-DOrSU-ODI-27			
Issue Status	Rev No.	Effective Date	Page No.
01	00	07.22.2022	2 of 2

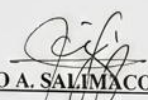
Certification of Statistical Analysis

This is to certify that the data presented in **"DORSU ALUMNI TRACER: A DATA-DRIVEN APPROACH TO MONITORING ALUMNI CAREER PROGRESS"** has undergone rigorous statistical analysis conducted by a certified statistician.

Certified Statistician:Rodrigo A. Salimaco Jr., PhD

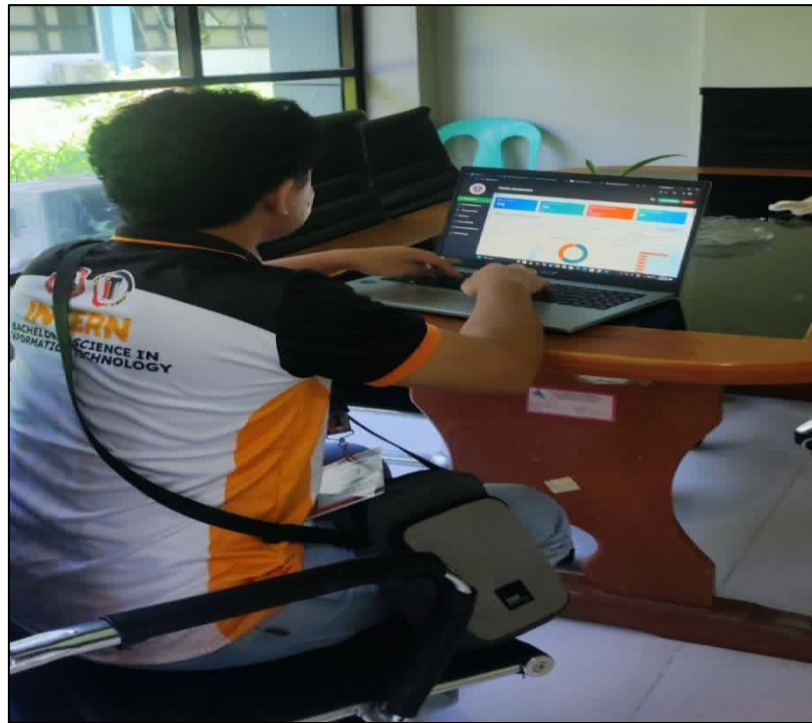
Date of Analysis: June 18, 2026

The statistical methods employed in this analysis are recognized and accepted within the statistical community for their accuracy and reliability.


RODRIGO A. SALIMACO JR., PhD
 Statistician

APPENDIX I.

User Testing Photo Documentation

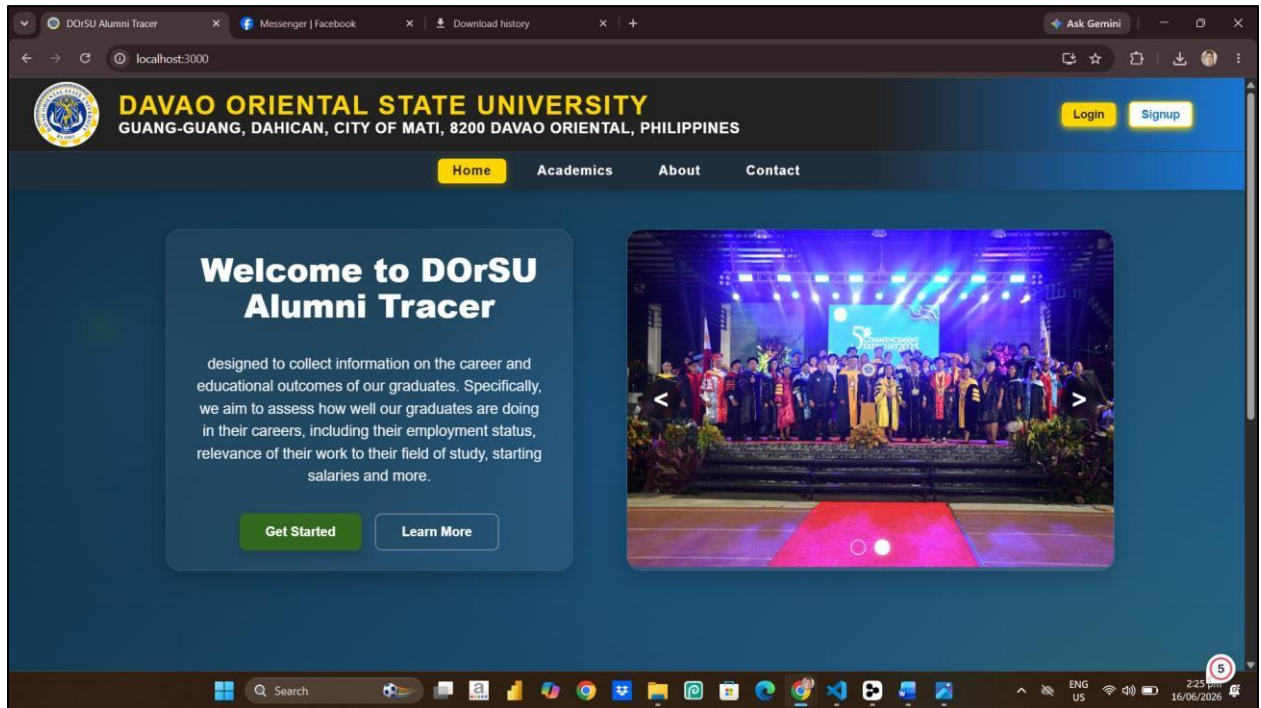




APPENDIX J.

DOrSU Alumni Tracer User Manual

1.0 System Overview



The DORSU Alumni Tracer System tracks graduate employment status for institutional research and CHED compliance.

User Roles:

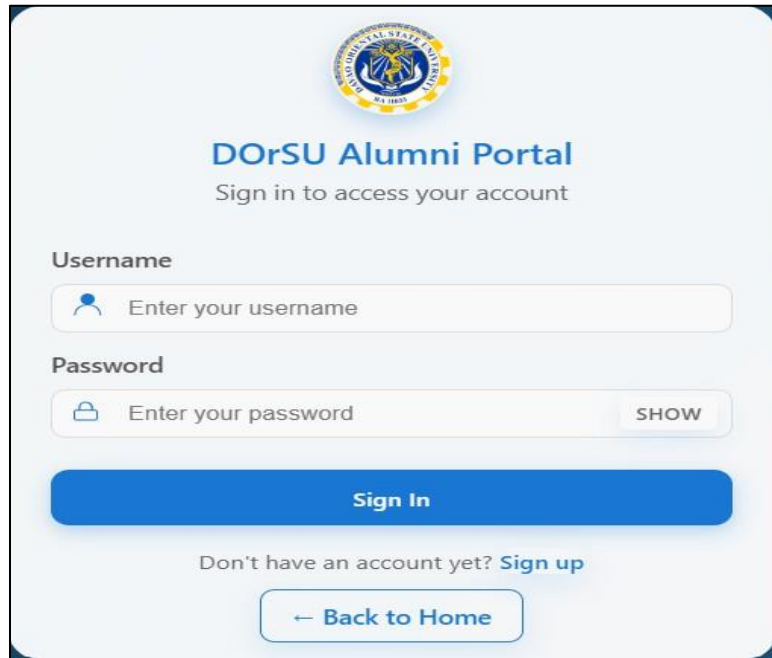
- Administrator: Full system control, user management, reports
- Alumni: Complete surveys, view job matching results
- Program Head: Monitor program-specific alumni data

System Requirements:

- Modern web browser (Chrome, Firefox, Edge)
- Internet connection
- Screen: 1024x768 minimum

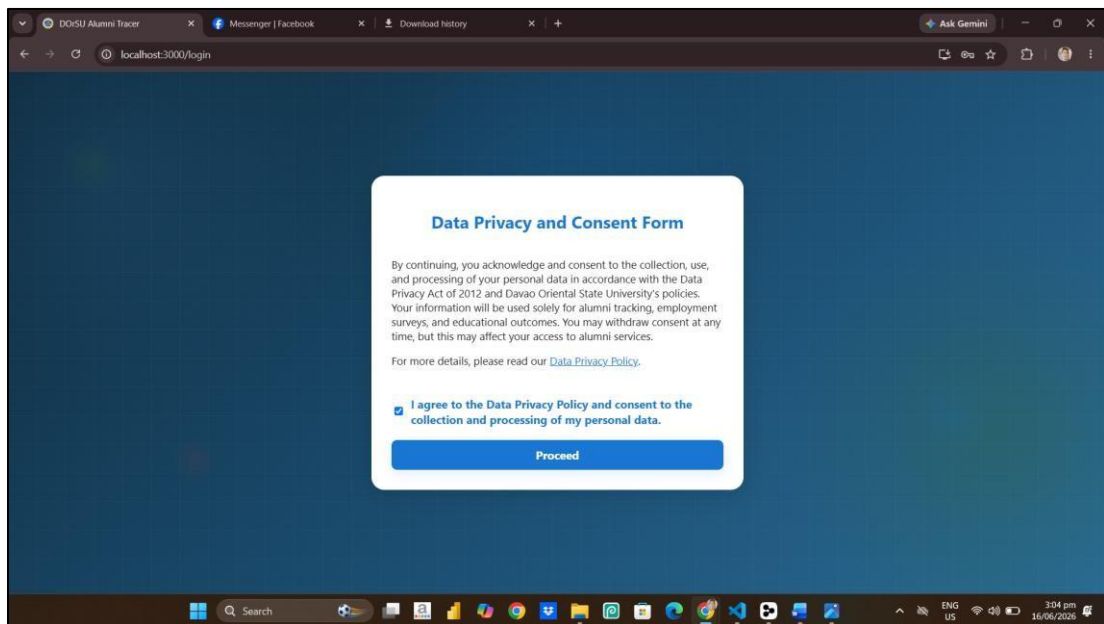
2.0 Getting Started

2.1 Login Credentials



The image shows the login page of the DOrSU Alumni Portal. At the top center is the Davao Oriental State University logo. Below it, the text "DOrSU Alumni Portal" is displayed in a large blue font, followed by "Sign in to access your account" in a smaller black font. The form contains two input fields: "Username" with a person icon and "Password" with a lock icon. A "SHOW" button is next to the password field. Below the inputs is a large blue "Sign In" button. At the bottom, there is a link "Don't have an account yet? Sign up" and a "← Back to Home" button.

2.2 Login Process

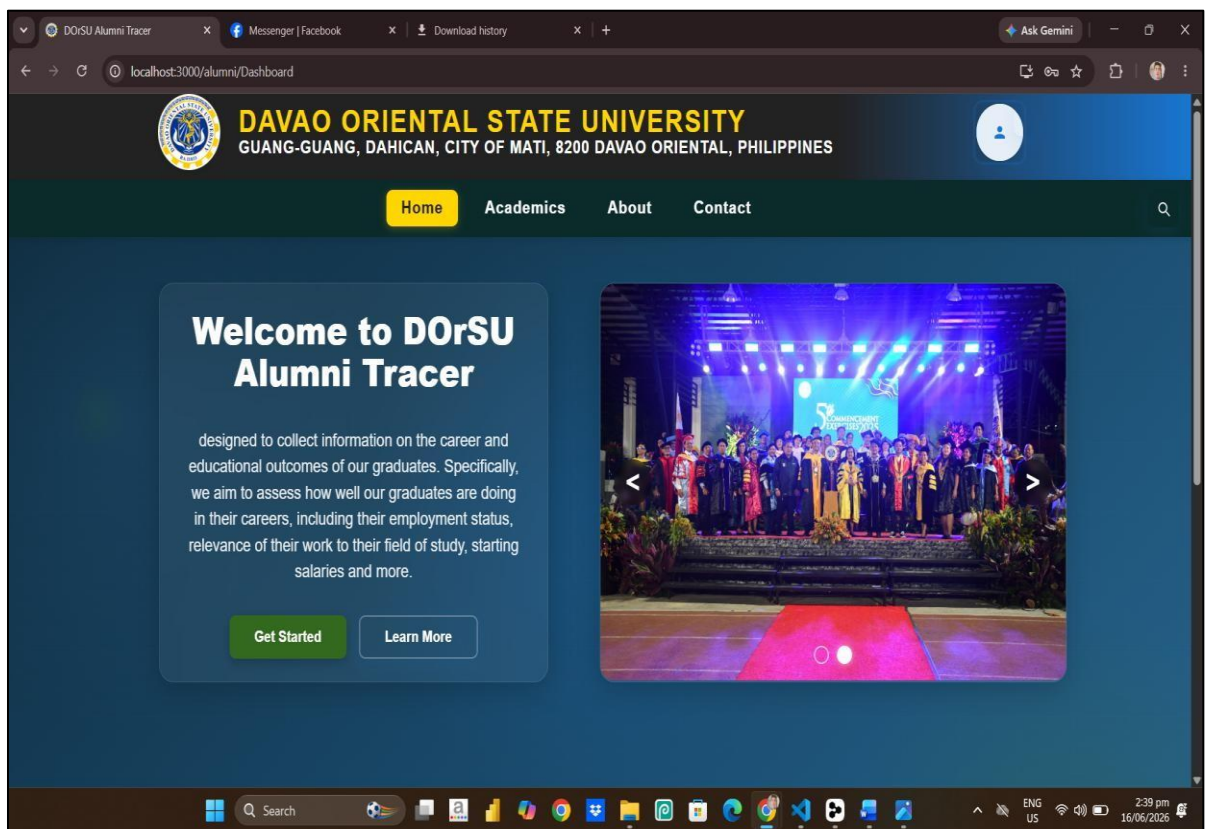


The image shows a web browser window displaying a "Data Privacy and Consent Form". The form is centered on a dark blue background. It contains the following text: "By continuing, you acknowledge and consent to the collection, use, and processing of your personal data in accordance with the Data Privacy Act of 2012 and Davao Oriental State University's policies. Your information will be used solely for alumni tracking, employment surveys, and educational outcomes. You may withdraw consent at any time, but this may affect your access to alumni services. For more details, please read our [Data Privacy Policy](#)." Below this text is a checkbox labeled "I agree to the Data Privacy Policy and consent to the collection and processing of my personal data." which is checked. At the bottom of the form is a blue "Proceed" button. The browser window shows multiple tabs, including "DOrSU Alumni Tracer", "Messenger | Facebook", and "Download history". The address bar shows "localhost:3000/login". The Windows taskbar at the bottom shows the time as 2:04 pm on 16/06/2026.

1. Go to the system URL
2. Click Login
3. Enter username and password
4. Complete Data Privacy Consent (first login only)
5. Access dashboard

3.0 Alumni Guide

3.1 Complete Graduate Tracer Survey



Path: Dashboard → Survey

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GUANG-GUANG, DAHICAN, CITY OF MATI, 8200 DAVAO ORIENTAL, PHILIPPINES

A. Socio-Demographic Profile

1. Name *

Last Name

Delta

First Name

Juan

Middle Name (Optional)

Cruz

2. Sex *

☒ Male

☐ Female

☐ Prefer not to say

DAVAO ORIENTAL STATE UNIVERSITY
GUANG-GUANG, DAHICAN, CITY OF MATI, 8200 DAVAO ORIENTAL, PHILIPPINES

A. Employment After Graduation

15. Were you Employed After Graduation? *

☒ Yes

☐ No

16. Was the Course degree you had in college relevant to your first Job? *

☒ Yes

☐ No

B. First Employment Details

17. Length of employment during your first employment: *

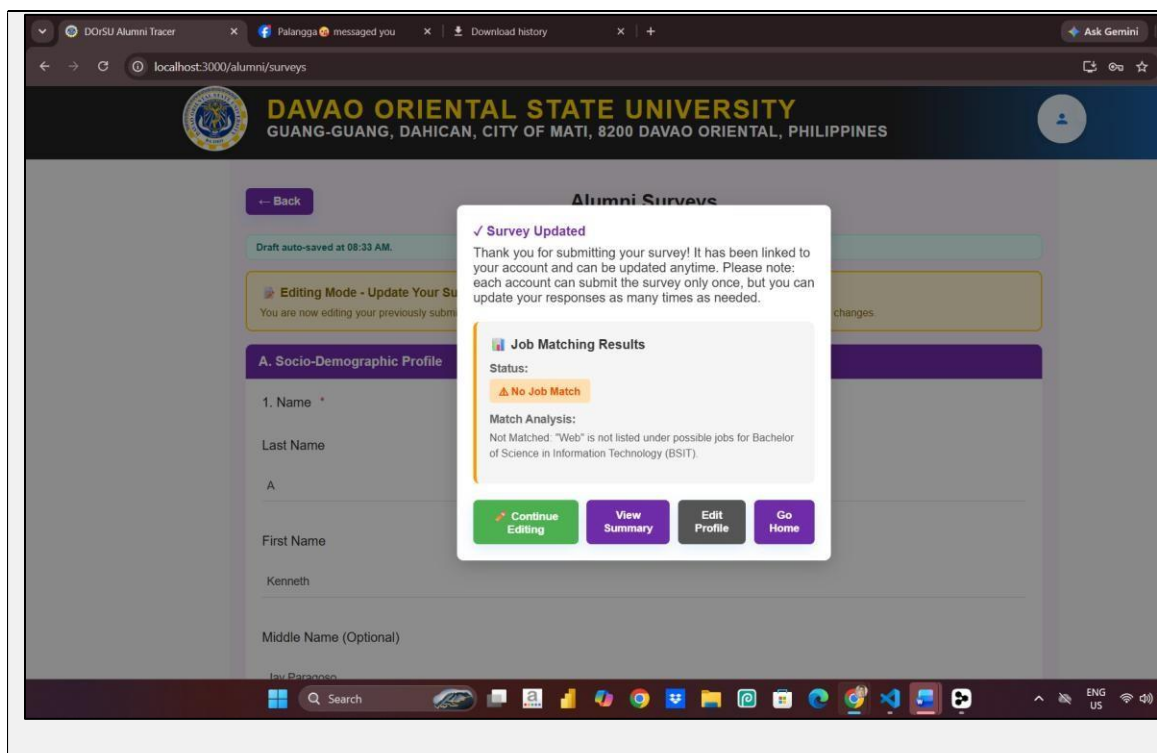
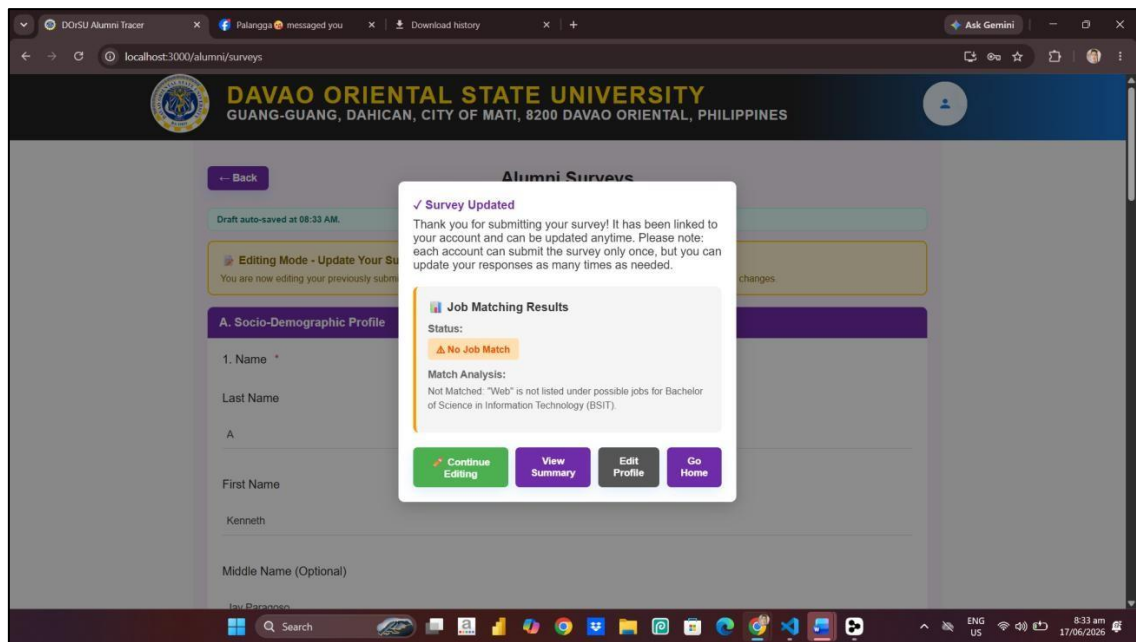
Choose

18. Sector of Employment in your first employment: *

Choose

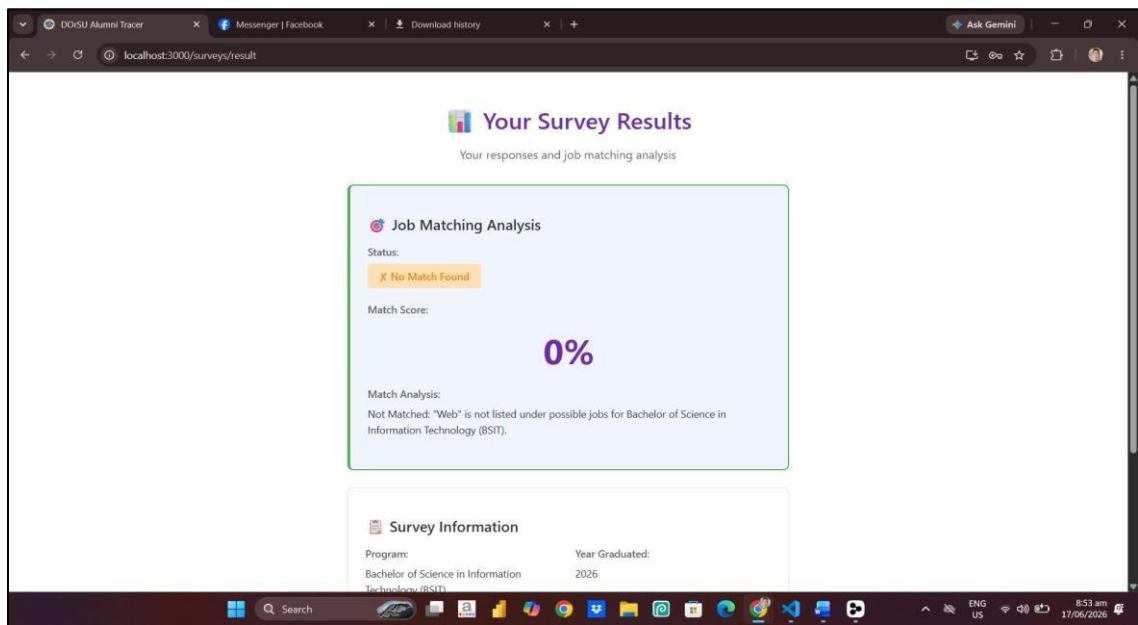
Fill all sections:

- Section A: Personal info (name, address, contact)
- Section B: Education (program, year graduated, honors)
- Section C: Employment (status, job title, company, salary)
- Section D: Job-course relationship
- Section E: Program evaluation (ratings, suggestions)



- Click Submit Survey
- View Job Matching Results immediately

3.2 View Job Matching Results



Job Matching Results			
TOTAL SURVEYS 3	MATCHED 2	NOT MATCHED 1	NOT APPLICABLE 0
Run Job Matching (All) Refresh			
CMO-Based Opportunities Select Program: -- Select a Program --			
Filter: All Surveys	Program: All Programs	Sort: Last Matched (Newest)	Search by name or ID...
Alumni Name	Program	Current Job Title	Match Status
Paragoso, Kei A.	Bachelor of Science in Information Technology (BSIT)	Back-end Developer	✓ Matched
Paragoso, Kenneth Jay A.	Bachelor of Science in Information Technology (BSIT)	Web Developer	✓ Matched

After survey submission, see:

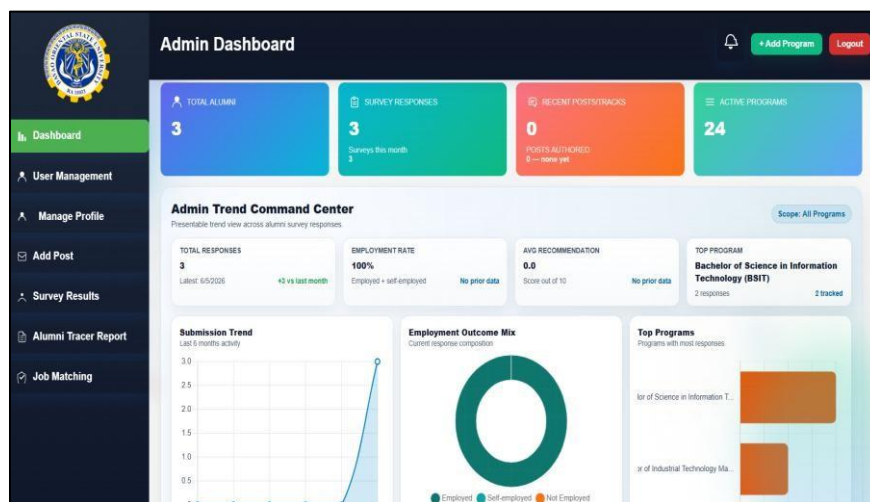
- Match Status: Match / Not Match / Not Applicable
- Match Score: 0-100%
- Reason: Explanation of result

- Top 3 Recommended Jobs: Course-appropriate roles

How It Works:

- Rule-based Algorithm with Fuzzy Matching
- Handles typos: "Developr" → "Developer" (95% match)
- Compares job title against CMO-prescribed job list

4.0 Administrator Guide

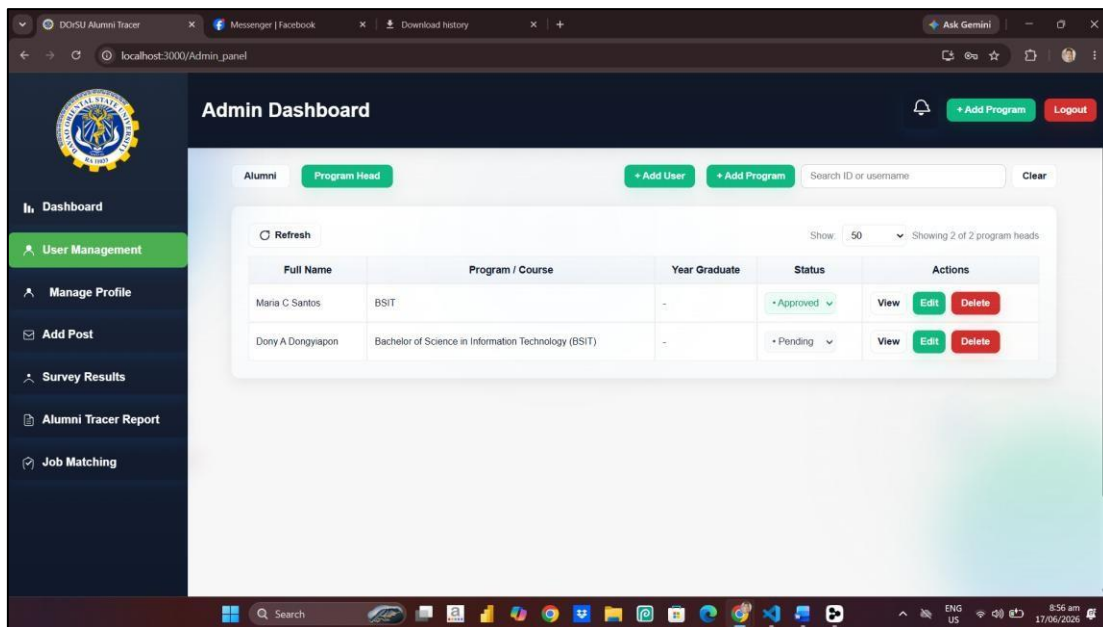


4.1 User Management

Path: Admin Dashboard → User Management

The User Management section allows administrators to manage the system's users. It includes a search bar, a refresh button, and a table of users with actions to view, edit, or delete each user.

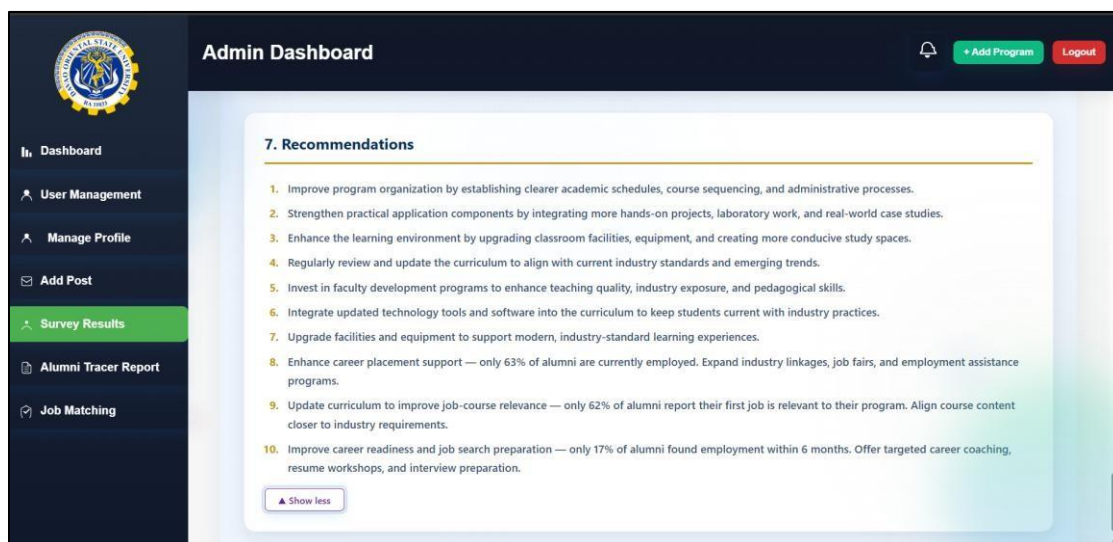
Full Name	Program / Course	Year Graduate	Actions
Dimaclangan, Marcelino Dandoy	Bachelor of Elementary Education (BEED)	2023	View Edit Delete
Santos, Ruperta Bondoc	Bachelor of Elementary Education (BEED)	2019	View Edit Delete
Accola, Ramon Garcia	Bachelor of Elementary Education (BEED)	2018	View Edit Delete
Daranglay, Encarnacion Cruz	Bachelor of Elementary Education (BEED)	2014	View Edit Delete
Calao, Consuelo Santos	Bachelor of Elementary Education (BEED)	2018	View Edit Delete
Reyes, Violacion Morales	Bachelor of Elementary Education (BEED)	2018	View Edit Delete
Darlagan, Pablo Dambao	Bachelor of Elementary Education (BEED)	2020	View Edit Delete
Dela Cruz, Florencio Dela Cruz	Bachelor of Elementary Education (BEED)	2020	View Edit Delete



- Alumni: View, edit, delete
- Program Heads: Approve/reject pending applications
- Search by name, email, or program

4.2 Survey Results

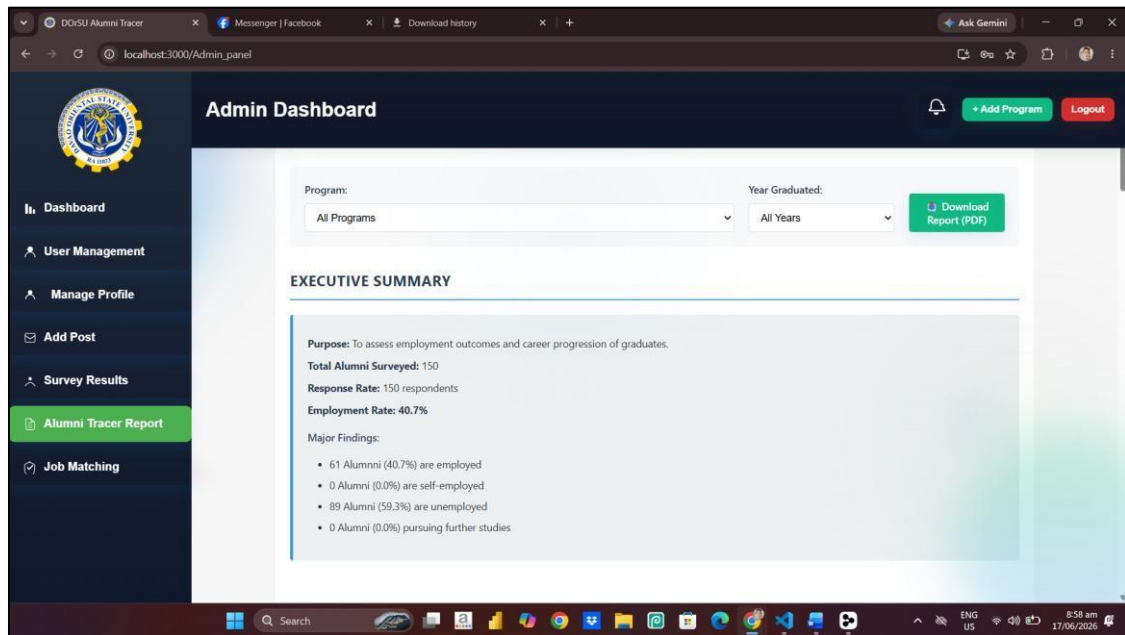
Path: Admin Dashboard → Survey Results



- View all submitted surveys in table format
- Filters: Program, Year, Employment Status.
- Click the individual survey for full details

4.3 Generate Tracer Report

Path: Admin Dashboard → Tracer Report



6. Set Date Range
7. Select Program Filter (optional)
8. Download file

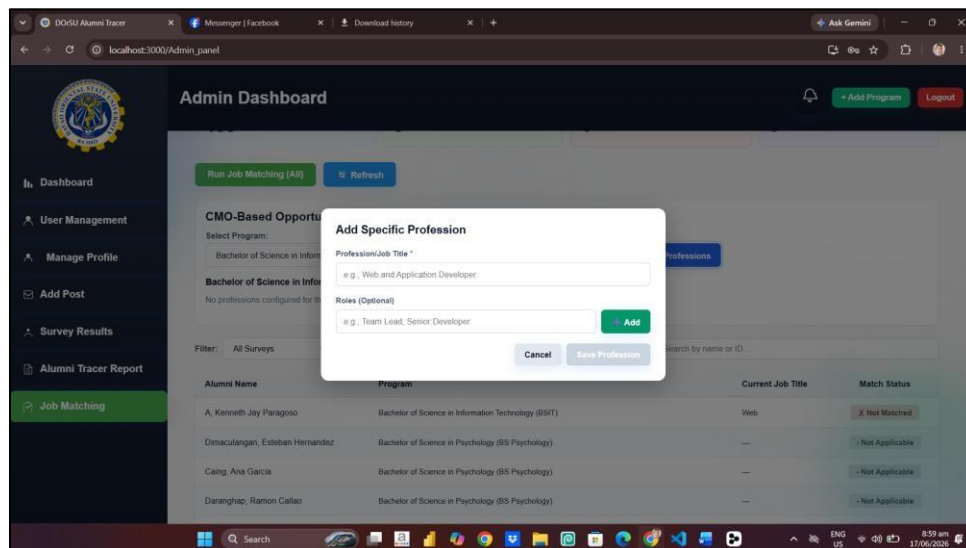
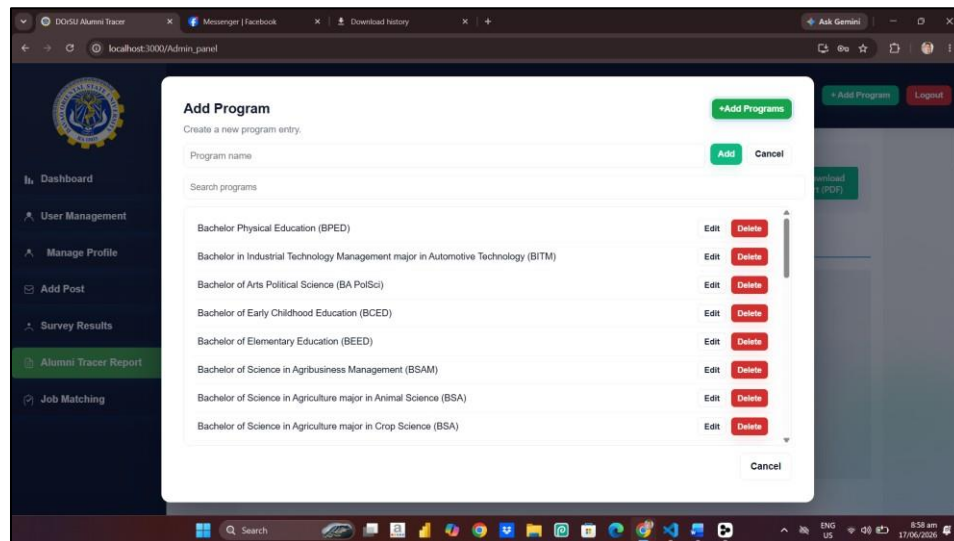
Report Contents:

- Employment status distribution
- Program-specific employment rates
- Job match analysis
- Salary distributions

- Time-to-employment metrics

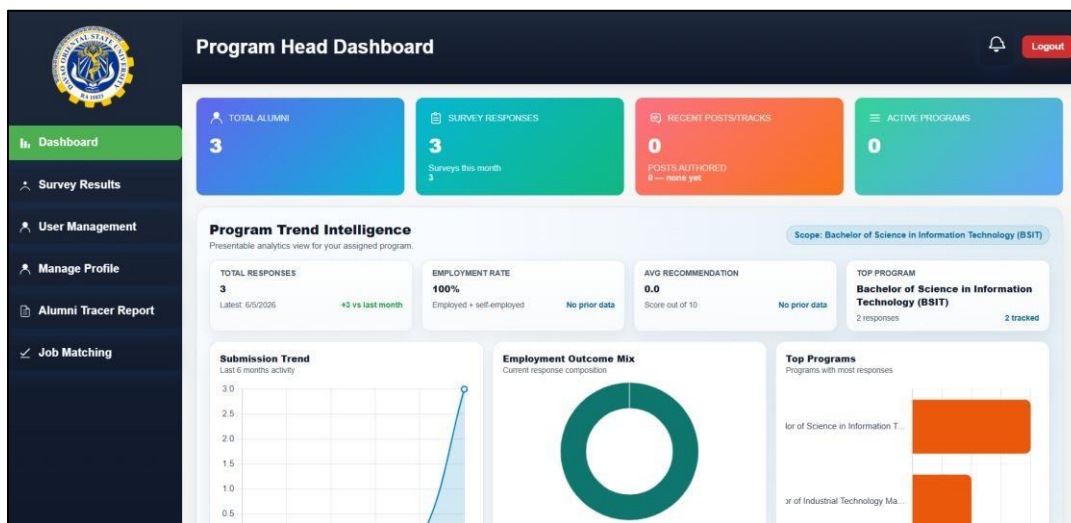
4.5 Program Management

Path: Admin Dashboard → Programs



- Add/Edit/Delete programs
- Manage Jobs: Add CMO-prescribed job opportunities per program

5.0 Program Head Guide



5.2 Alumni List

Path: Program Head Dashboard → Alumni List

The screenshot shows the 'Alumni List' page, filtered for the Bachelor of Science in Information Technology (BSIT) program. The table lists graduates with their full names, program/course, year of graduation, and actions (View, Edit, Delete). The table is sorted by year of graduation in ascending order.

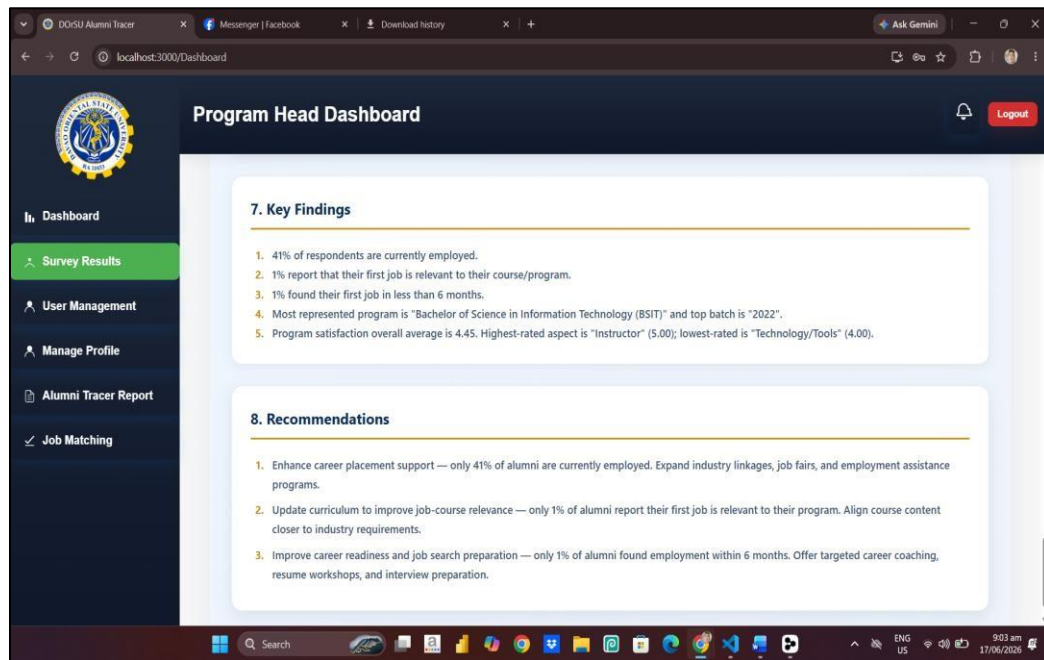
Full Name	Program / Course	Year Graduate	Actions
Hernandez, Policarpo Dela	Bachelor of Science in Information Technology (BSIT)	2019	View Edit Delete
Danilo, Ortelia Darang	Bachelor of Science in Information Technology (BSIT)	2021	View Edit Delete
Camiling, Hermenegildo Daraza	Bachelor of Science in Information Technology (BSIT)	2021	View Edit Delete
Darahap, Hermenegildo Reyes	Bachelor of Science in Information Technology (BSIT)	2023	View Edit Delete
Dapuran, Saturnino Cabrera	Bachelor of Science in Information Technology (BSIT)	2023	View Edit Delete
Torres, Eulalia Reyes	Bachelor of Science in Information Technology (BSIT)	2024	View Edit Delete
Caing, Hermenegildo Camiling	Bachelor of Science in Information Technology (BSIT)	2018	View Edit Delete
Daranghap, Vicente Ramos	Bachelor of Science in Information Technology (BSIT)	2014	View Edit Delete
Daraus, Martina Bondoc	Bachelor of Science in Information Technology (BSIT)	2024	View Edit Delete

- View program-specific alumni

- Filters: Graduation Year, Survey Status, Employment Status

5.3 Survey Analytics

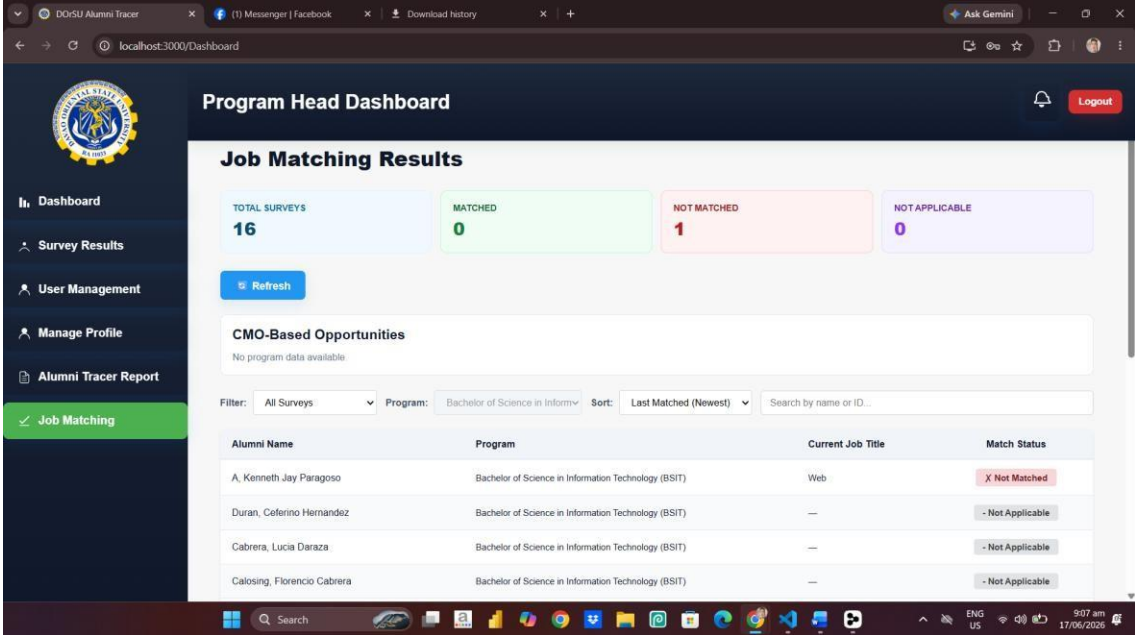
Path: Program Head Dashboard → Survey Analytics



- Employment status distribution
- Salary range breakdown
- Job sector distribution
- Satisfaction ratings
- Export program reports

5.4 Job Matching View

Path: Program Head Dashboard → Job Matching



Program Head Dashboard

Job Matching Results

TOTAL SURVEYS: 16 | MATCHED: 0 | NOT MATCHED: 1 | NOT APPLICABLE: 0

[Refresh](#)

CMO-Based Opportunities
No program data available.

Filter: All Surveys | Program: Bachelor of Science in Inform | Sort: Last Matched (Newest) | Search by name or ID...

Alumni Name	Program	Current Job Title	Match Status
A, Kenneth Jay Paragoso	Bachelor of Science in Information Technology (BSIT)	Web	X Not Matched
Duran, Ceferino Hernandez	Bachelor of Science in Information Technology (BSIT)	—	- Not Applicable
Cabrera, Lucia Daraza	Bachelor of Science in Information Technology (BSIT)	—	- Not Applicable
Calosing, Florencio Cabrera	Bachelor of Science in Information Technology (BSIT)	—	- Not Applicable

- View course-specific keywords
- Understand matching criteria

CURRICULUM VITAE



KRISTINE MAE GONZALES BONOTAN

A BSIT student at Davao Oriental State University who demonstrates leadership, initiative, and a strong sense of responsibility. Committed to continuous personal and professional growth, with the ability to manage tasks efficiently and work well under pressure.

📞 09264283871
 ✉ kristinebonotan12@gmail.com
 📍 Mati City, Davao Oriental

Personal Background

Age: 22 **Gender:** Female
Birthday: December 02, 2003
Height: 5'1 ft **Weight:** 42 kg
Religion: Catholic
Status: Single

Educational Background

2010 - 2011	Mayor Luisito Garcia Rabat Memorial School (MLGR)
2021 - 2022	Mati National Comprehensive High School (MNCHS)
2025 - 2026	Davao Oriental State University - Main Campus (DOrSU)

Skills & Interests

- Data Encoding and Record Keeping
- Inventory Monitoring and Stock Management
- Document Processing and Filing
- Microsoft Office Suite (Excel, Word, PowerPoint)
- Photo editing (Canva)
- Problem Solving
- Teamwork and Collaboration



MIKE ALBACITE MISOLES

Bachelor of Science in Information Technology (BSIT) student with knowledge of programming, web development, and computer systems. Passionate about creating efficient technology solutions and expanding technical expertise through practical experience and continuous learning.

📞 09554584942
 📧 mikemisoles74@gmail.com
 📍 Mati City, Davao Oriental

Personal Background

Age: 22 **Gender:** Male

Birthday: March 30, 2004

Height: 5'5 ft **Weight:** 59 kg

Religion: Catholic

Status: Single

Educational Background

2010
-
2011

Marayag Elementary School

2021
-
2022

**Mati National Comprehensive
High School (MNCHS)**

2025
-
2026

**Davao Oriental State University
- Main Campus (DORSU)**

Skills & Interests

- Data Encoding and Record Keeping
- Administrative and Clerical Operations
- Inventory Monitoring and Stock Management
- Customer Service and Client Assistance
- Document Processing and Filing
- Customer Service
- Microsoft Office Suite (Excel, Word, PowerPoint)
- UI/UX Design
- Data Entry
- Events Spinner
- Web Development



KENNETH JAY PARAGOSO

A BSIT student at Davao Oriental State University with a passion for technology, innovation, and problem-solving. Committed to continuous improvement and capable of working independently or collaboratively to achieve organizational goals while developing professional IT competencies.

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 ✉️ paragosokennethjay@gmail.com
 📍 Marayag Lupon, Davao Oriental

Personal Background

Age: 22 **Gender:** Male

Birthday: November 07, 2003

Height: 5'5 ft **Weight:** 58 kg

Religion: UPC

Status: Single

Educational Background

2010 - 2011	Marayag Elementary School
2021 - 2022	Marayag High School
2025 - 2026	Davao Oriental State University - Main Campus (DORSU)

Skills & Interests

- Multitasking
- Inventory Monitoring and Stock Management
- Document Processing and Filing
- Photo editing (Canva)
- Web Development
- Technology
- Encoding data